VAJIRAM & RAVI

Institute for IAS Examination

PowerUp

Current Affairs Test Series 2025

CA Test – 09 – (CA2259)

Answer Key
&
Detailed Answer Explanations

CA Test - 09 - (CA2259) - Answer Key									
1. (a)	11. (a)	21. (c)	31. (d)	41. (c)	51. (a)	61. (c)	71. (c)	81. (d)	91. (b)
2. (c)	12. (c)	22. (a)	32. (b)	42. (c)	52. (b)	62. (c)	72. (d)	82. (b)	92. (a)
3. (a)	13. (d)	23. (d)	33. (c)	43. (b)	53. (b)	63. (b)	73. (b)	83. (a)	93. (d)
4. (b)	14. (a)	24. (a)	34. (b)	44. (c)	54. (a)	64. (d)	74. (d)	84. (c)	94. (a)
5. (d)	15. (a)	25. (b)	35. (d)	45. (d)	55. (a)	65. (b)	75. (b)	85. (a)	95. (d)
6. (a)	16. (c)	26. (b)	36. (c)	46. (b)	56. (b)	66. (b)	76. (d)	86. (a)	96. (a)
7. (c)	17. (b)	27. (c)	37. (c)	47. (b)	57. (b)	67. (a)	77. (b)	87. (a)	97. (a)
8. (b)	18. (d)	28. (a)	38. (b)	48. (c)	58. (b)	68. (c)	78. (b)	88. (c)	98. (c)
9. (d)	19. (a)	29. (c)	39. (c)	49. (c)	59. (d)	69. (a)	79. (b)	89. (c)	99. (b)
10. (b)	20. (d)	30. (a)	40. (b)	50. (d)	60. (a)	70. (a)	80. (a)	90. (d)	100. (c)

Q1.
Answer: a
Explanation:

- **Dark comets** are celestial bodies which **lack the glowing tails** (a typical feature of comets) and hence, resemble asteroids.
- They have less surface area for material to escape and form into the beautiful tails we see on typical comets. But **they are not asteroids due to their sudden accelerations**, which suggests that they may be firing jets of gas just like comets.
- They often spin quite rapidly and disperse escaping gas and dust in all directions, making them less visible.
- They follow **elongated, elliptical paths** that bring them close to the Sun before sweeping back out to the farthest reaches of the Solar System.
- The first indication of dark comets came in 2016, when asteroid 2003 RM exhibited unusual orbital deviations. Since then, astronomers have confirmed the existence of dark comets, with a new study published which revealed 14 such objects.
- These comets fall into two main categories: "**outer dark comets," which have eccentric orbits** and are **larger, and "inner dark comets**," which are smaller and closer to the Sun, with nearly circular orbits.

Therefore, option (a) is the correct answer.
Relevance: The National Aeronautics and Space Administration (NASA) has announced the discovery of seven new 'dark comets' taking the total number to 14.
Subject: Science & Technology | Physics
Tags: Conceptual, Factual

Q2.
Answer: c
Explanation:

- **Forest composition r**efers to all plant species found in a stand or landscape, including trees, shrubs and grasses.

- Recent research has found that **climate change is altering the tree line landscape of the central Himalayas. So, statement 1 is correct.**
- According to the research, **the change is due to replacement of birch trees by fir trees.** Birch trees is a **deciduous broadleaved species** that dominates the region. They are being replaced by fir, **a slow-growing evergreen conifer**. This shift is likely to alter the ecological function of Himalayan ecosystems.
    - Fir trees (*Abies spectabilis*) are typically found at a lower altitude compared to the Himalayan birch (*Betula utilis*), but the research noted a shift in the pattern. While both species are climbing upslope on the highlands, **fir is climbing faster.**
- According to the study, the reason is the warming of climate. The central Himalayas are experiencing rapid climatic changes. **The conifers are likely benefiting from a warmer climate while deciduous birch trees are struggling to grow due to stress from increased temperatures and lack of moisture**. Scientists found that fir trees have been growing more in recent decades, likely because they benefit from warmer temperatures. In contrast, birch trees, which need more moisture, are growing less due to heat-related dryness. **So, statement 2 is not correct.**
- Over the past 200 years, fir has climbed at an average of 11 centimetres (cm) per year, while birch has moved at a slower pace of 6 cm annually. The scientists estimate that the fir will outcompete birch at mixed tree lines, eventually becoming the dominant tree line species in the Himalayas.
- Other factors, like **fewer birch seedlings** and interactions with other plants, are also causing this decline.

So, Statement–I is correct, but Statement–II is incorrect.
Therefore, option (c) is the correct answer.
Relevance: A new recent research has found that Climate change is altering the tree line landscape of the central Himalayas.
Subject: Environment | Climate Change
Tags: Factual, Conceptual

Q3.
Answer: a
Explanation:

- **PS4-Orbital Experiment Module**, designated as **POEM-4**, refers to the **usage of the spent fourth stage** of the Polar Synchronous Launch Vehicle (PSLV). **PSLV-C60 SpaDeX mission** is the fourth POEM Mission (in short POEM-4) in the series.
- A total of **24 payloads** will be flown in this POEM-4 mission, of which **14 payloads** are from ISRO-DoS centres and **10 payloads** are from various **Non-Government Entities (NGEs)** comprising Academia and Start-ups that have been received through **IN-SPACe.**
- Key highlights of the POEM-4 payloads include **robotic experiments** such as the Walking Robotic Arm and Debris Capture Manipulator, advanced sensor technologies like the **Reaction Wheel Assembly** and **Multi-Sensor Inertial Reference System** and green propulsion systems.
- It also features space science experiments like the Electron Temperature Analyzer and **AI-based Earth observation payloads,** enhancing India's space research capabilities.

Therefore, option (a) is the correct answer.

Relevance: The Indian National Space Promotion and Authorization Centre (IN-SPACe) recently announced the successful deployment of 10 hosted payloads by Non-Governmental Entities (NGEs) on ISRO's PSLV-C60/SpaDeX mission using the POEM-4 module.
Subject: Science & Technology | Space and Related Developments
Tags: Factual

Q4.
Answer: b
Explanation:

- **Geo-heritage refers** to the geological features that are important either because of their **natural significance** or their **cultural value.**
- The **Geological Survey of India (GSI)** is the main organization **responsible for identifying** and preserving geo-heritage sites and national geological monuments throughout the country. **So, statement 1 is not correct.**
- Protection and maintenance of geo-heritage sites involve measures by both the GSI and the respective State governments. While GSI declares the sites, **State governments implement necessary policies**, **infrastructure**, and **protective measures** to preserve them. **So, statement 2 is correct.**
- **Rajasthan has the highest number** of geo-heritage sites in India, with **12 out of the total 34** geo-heritage sites recognized by the Geological Survey of India (GSI). Notable locations include the **Stromatolite Fossil Park** in Jhamarkotra, Akal Fossil Wood Park in Jaisalmer, and the Great Boundary Fault at Satur etc. **So, statement 3 is correct.**

Therefore, option (b) is the correct answer.

Knowledge Box
34 Geo-Heritage Sites in India: ● **Andhra Pradesh** ○ Volcanogenic Bedded Barytes, Cuddapah District ○ Eparchaean Unconformity, Anantapur District ○ Natural Geological Arch, Tirumala Hills, Chittoor District ○ Erra Matti Dibbalu, Visakhapatnam District ● **Kerala** ○ Laterite, Malappuram District ○ Varkala Cliff Section, Thiruvananthapuram District ● **Tamil Nadu** ○ Fossil Wood Park, Villupuram District ○ National Fossil Wood Park, Perambalur District ○ Charnockite, St. Thomas Mount, Chennai ○ Badlands of Karai Formation with Cretaceous Fossils, Karai, Perambalur District ● **Gujarat** ○ Sedimentary Structures Eddy Markings, Kadan Dam, Panch Mahals District ● **Rajasthan** ○ Sendra Granite, Pali District ○ Barr Conglomerate, Pali District

- Stromatolite Fossil Park, Jhamarkotra Rock Phosphate Deposit, Udaipur District
- Gossan in Rajpura-Dariba Mineralised Belt, Udaipur District
- Stromatolite Park, Chittorgarh District
- Akal Fossil Wood Park, Jaisalmer District
- Kishangarh Nepheline Syenite, Ajmer District
- Welded Tuff, Jodhpur District
- Jodhpur Group – Malani Igneous Suite Contact, Jodhpur District
- Great Boundary Fault at Satur, Bundi District
- Ramgarh Crater, Baran District
- Zawar Lead-Zinc Mine, Udaipur District

- **Maharashtra**
  - Lonar Lake, Buldana District
- **Chhattisgarh**
  - Lower Permian Marine Bed, Manendragarh, Sarguja District
- **Karnataka**
  - Columnar Basaltic Lava, St. Mary Islands, Udupi District
  - Pillow Lavas, Chitradurga District
  - Peninsular Gneiss, Bangalore
  - Pyroclastics & Pillow Lavas, Kolar Gold Fields, Kolar District
- **Himachal Pradesh**
  - Siwalik Fossil Park, Sirmur District
- **Odisha**
  - Pillow Lava in Iron Ore Belt, Keonjhar District
- **Jharkhand**
  - Plant Fossil Bearing Inter-trappean Beds of Rajmahal Formation (Upper Gondwana Sequence), Mandro, Sahibganj District
- **Nagaland**
  - Nagahill Ophiolite – Site near Pungro
- **Sikkim**
  - Stromatolite Bearing Dolomite/Limestone of Buxa Formation, Mamley, near Namchi, South District

Relevance: The National Green Tribunal has sought a status report on Kerala's Varkala Cliff, a geo-heritage site facing degradation due to unregulated tourism and coastal erosion.
Subject: Geography | Environment
Tags: Factual

Q5.
Answer: d
Explanation:

- **CROPS** stands for Compact Research Module for Orbital Plant Studies which is an unmanned experimental module designed to develop capabilities for growing and sustaining plants in space. The experiment is being performed by the Indian Space Research Organisation (ISRO).

- CROPS first mission (CROPS – 1) is designed to demonstrate **germination of a seed and growth of plant up to two leaves stage in space.** It is an airtight container of diameter 300mm and height 450mm simulating an earth like environment in space except gravity.
- Growing plants from seed in space under microgravity is an important milestone for space biological research, to produce food in space and as an engaging activity **for the astronauts in space during long duration manned missions**.
- **Cowpea (*Vigna unguiculata*) was chosen for the CROPS**, based on ground trials on a variety of seeds. ISRO selected cowpea over other plants due to its **relatively fast growth rate and robust root and stem structure**. It can grow to a height of, say, 100 mm or more reaching the two-leaf stage in five to seven days.
- **Cowpea** is a leguminous crop having long trailing vines producing seeds of black, white, or red colour used widely in various food preparations. It originated and was domesticated in Southern Africa and was later moved to East and West Africa and Asia. The plant **tolerates drought, and performs well in a wide variety of soils**. However, **it is not the correct reason** for being chosen by the ISRO.
- The world's **largest production and consumption of cowpea is in West Africa**, which accounts for more than 87% of world production and use. Nigeria, the largest producer and consumer, accounts for 46% of production worldwide.
- It is a minor pulse in India, cultivated mainly in arid and semi-arid tracts.

Therefore, option (d) is the correct answer.
Subject: Science & Technology | Space and Related Developments
Tags: Conceptual, Factual

Q6.
Answer: a
Explanation:

- The **Future Skills Report** is a report that assesses how well countries are preparing their job markets for future demands. The report evaluates factors such as the availability of skilled workers, the effectiveness of skill development programs, and future job creation. It is published by the **UK-based higher education firm QS**. Ranked 25th overall, India is categorised as a 'future skills contender' in the index.**So, pair 1 is not correctly matched.**
- The **Future of Jobs Report** explores how jobs and skills will evolve in future. The report is based on unique survey data that details the expectations of a cross-section of the world's largest employers related to how socio-economic and technology trends will shape the workplace of the future. It is published by the World Economic Forum. **So, pair 2 is not correctly matched.**
- The **International Labour Organization** (ILO) releases the **World Employment and Social Outlook** (WESO) report. The report analyzes the state of the global labor market, including trends and challenges. The report provides a detailed analysis of the state of global labour markets, highlighting key trends and challenges. It explores issues such as the slowdown in labour market recovery, the persistence of youth unemployment, and the widening global jobs gap. Additionally, the report examines the impact of geopolitical tensions, climate change, and economic inequalities on labour markets. **So, pair 3 is correctly matched.**

So, only one of the pairs given above is correctly matched.
Therefore, option (a) is the correct answer.

Relevance: The International Labour Organization (ILO) published its World Employment and Social Outlook: Trends 2025 (WESO Trends) recently.
Subject: Economics | Economic Organisations and Conventions
Tags: Factual

Q7.
Answer: c
Explanation:

- The **MSE-SPICE Scheme** is the "**MSE Scheme for Promotion and Investment in Circular Economy**" introduced by the Ministry of Micro, Small & Medium Enterprises (MoMSME). It aims to encourage Micro, Small, and Medium Enterprises (MSEs) to **adopt circular economy practices,** focusing on sectors like plastic, rubber, and electronics waste management to comply with **international environmental goals** and improve operational efficiency.
- The categories of enterprises covered under the MSE-SPICE Scheme are **Micro and Small Enterprises (MSEs)** that are registered with the **UDYAM portal** of the Ministry of Micro, Small & Medium Enterprises.
- **The key components** of the MSE-SPICE Scheme are as follows:
    - Credit Linked Capital Subsidy Component
    - Awareness Generation and Demand Creation Component
    - Information, Education, and Communication Component (IEC).

Therefore, option (c) is the correct answer.
Relevance: As of December 12, 2024, under the MSE-SPICE scheme, six MSE accounts have been assisted, with an expenditure of ₹.31 crore out of a ₹72.5 crore outlay.
Subject: Economics | Sustainable Development and Growth
Tags: Factual

Q8.
Answer: b
Explanation:

- The **Assessment Report on the Interlinkages Among Biodiversity, Water, Food and Health** is also known as the Nexus Report.
- It is released **by the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES). So, statement 1 is not correct.**
- The report **discusses the impact of climate change on biodiversity, human health and well-being.** The report says that governments are underestimating or ignoring the links between five key areas - biodiversity, water, food, health and climate change. It highlights the strong interconnections between global challenges such as **climate change, biodiversity loss, hunger, water scarcity, and health risks**. It emphasizes that tackling these issues separately is not only ineffective but also counterproductive, as they interact and compound each other. It emphasizes tackling these issues in cohesion. The report is **not a ranking exercise**. **So, statement 2 is not correct but statement 3 is correct.**
- The recent report held that the **heat stress and lack of water availability would affect crop yield**, in turn impacting terrestrial food production.

- It also stated that in the past 50 years, extreme weather, climate- and water-related events have caused nearly 12,000 disasters, leading to 2 million human deaths. 90% of these were reported from low- and lower-middle-income countries.
- The report also found more than half of the world's population live in areas hit by the biggest impacts from declines in biodiversity, water and food.

Therefore, option (b) is the correct answer.

Relevance: Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES) released the Nexus Report recently.

Subject: Environment | Environmental Organisations and Conventions

Tags: Factual

Q9.

Answer: d

Explanation:

- **Patola is a double ikat woven sari made in Patan, Gujarat.** It is usually made from silk. They are very expensive, and once worn only by those belonging to royal and aristocratic families. Velvet patola styles are also made in Surat. Patola-weaving is a closely guarded family tradition. **So, pair 1 is not correctly matched.**
- **Jamdani is centuries-old handwoven textile from the Bengal region, now Bangladesh and West Bengal.** Jamdani Saree is one of the finest handwoven saree of Bengal**.** Jamdani is a fine muslin cotton textile. In 2013, the traditional art of weaving jamdani was declared a UNESCO Intangible Cultural Heritage of Humanity. **So, pair 2 is correctly matched.**
- **Moirang Phee is a handwoven fabric and saree that originated in the Moirang village of Manipur.** The design of Moirang Phee is said to represent the thin and pointed teeth of Pakhangba, the pythonic god of Manipur mythology. The fiber used to make the yarn is derived from "lashing" (cotton ball) and "labrang" (mulberry silk). **So, pair 3 is correctly matched.**
- **Pochampally ikat is a saree made in Telangana.** Pochampally Ikat received a Geographical Indicator (GI Status) in 2004. Sarees are woven through a unique style called Ikat. Ikat involves the process of wrapping (or tying) and dyeing sections of bundled yarn to a predetermined colour pattern before they are woven. The dye penetrates into exposed sections while the wrapped section remains undyed. This pattern formed by the yarn in this process is woven into fabric. **So, pair 4 is correctly matched.**
- The **tangail sarees** are woven in **Nadia and Purba Bardhhaman district of West Bengal.** The hugely popular tangail cotton sarees have finer count and are decorated with extra warp designs using coloured yarn. It is a simplification of the jamdani cotton saree but with minimal designs in the body portion. It has recently received a GI Tag. **So, pair 5 is not correctly matched.**

Therefore, option (d) is the correct answer.

Relevance: The Ministry of Textiles organised the third edition of the Mega event "Viraasat Sari Festival 2024", from 15th – 28th December 2024, at Handloom Haat, Janpath, New Delhi.

Subject: Geography | Indian Social and Economic Geography

Tags: Factual

Q10.
Answer: b
Explanation:

- **The Schengen Area eliminates internal border checks**, facilitating free and safe movement for European Union (**EU) citizens, non-EU residents** and **visitors**. This area includes **most EU States and some associated non-EU countries**, creating a seamless travel experience. Border checks are only conducted at the external borders of the Schengen Area. **So, statement 1 is correct.**
- Participation in the Schengen Area is **not mandatory for all EU member states.** For example, **Ireland opted out of the Schengen Area**, and certain **non-EU countries** like **Norway, Switzerland, and Iceland are part** of it through agreements. Thus, the membership in the EU and participation in the Schengen Area are separate. **So, statement 2 is not correct and statement 3 is correct.**
- **Not all non-EU citizens** require a **Schengen visa** to enter the Schengen Area. The EU has a list of countries whose citizens are **exempt from requiring a visa** for short stays (up to 90 days) and those who must obtain a Schengen visa. For example, citizens from the **U.S., Canada, and Australia** are visa-exempt for short visits. **So, statement 4 is not correct.**

So, only two of the statements given above are correct.
Therefore, option (b) is the correct answer.
Relevance: The European Union has approved Romania and Bulgaria full entry into the Schengen Zone via land borders starting January 1, 2025.
Subject: Polity | International Relations
Tags: Factual

Q11.
Answer: a
Explanation:

- **Environmental factors** such as **air pollution and road traffic noise** have emerged as significant contributors to **infertility in both men and women.** The environmental stressors disrupt hormonal balance, induce oxidative stress, and impair reproductive functions, making them critical concerns for public health. **So, statement 1 is correct.**
- **PM2.5 particles** (particulate matter less than 2.5 micrometers in diameter) contain toxic elements like **heavy metals and volatile organic compounds (VOCs)** that generate reactive oxygen species **(ROS), causing oxidative stress.** This can **damage sperm DNA, reduce sperm motility, and disrupt testosterone production, leading to male infertility**. PM2.5 also disrupts hormone production, particularly testosterone, which is crucial for sperm formation and male reproductive health. **So, statement 2 is correct.**
- Chronic exposure to **road traffic noise at levels above 55 decibels** can increase **cortisol** (stress hormone) levels, disrupting **estrogen and progesterone balance**. Noise pollution can also trigger oxidative stress and inflammation, affecting egg quality and implantation in the uterus. S**o, statement 2 is correct.**
- This can result in menstrual irregularities, including:
    - **Anovulation (absence of ovulation**) – preventing egg release and reducing fertility.
    - **Amenorrhea (absence of menstruation**) – affecting reproductive health.

- **Noise pollution** raises cortisol levels, which can **negatively impact reproductive health.** High cortisol levels can **reduce testosterone production,** leading to lower sperm count and quality**.**

So, both Statement–II and Statement–III are correct and both of them explain Statement–I. Therefore, option (a) is the correct answer.
Relevance: A recent study published in the BMJ Journal highlights the link between long-term exposure to PM2.5 and road traffic noise to infertility in men and women.
Subject: Environment | Pollution
Tags: Factual, Analytical

Q12.
Answer: c
Explanation:

- **Risa is Tripura's traditional attire**. It has been given a Geographical Indication (GI) tag. Risa is known for its surprising and stylish designs, exclusive multi-colour combinations, and lasting texture. The Tripura tribal women make all the fabrics, including Risa, with the help of loin looms. They make Risa with multi-coloured warp and weft threads on the loom and create the most amazing, stylish designs. **So, pair 1 is correctly matched.**
- **Lakadong turmeric is a special variety of turmeric grown in Meghalaya.** It is considered one of the best in the world due to its **medicinal properties,** mainly **curcumin** content of up to 12%. The curcumin content of other turmeric varieties is 2-3%**. Lakadong received the GI tag in 2023. So, pair 2 is correctly matched.**
- **The traditional Assamese Gamosa/Gamusa or Gamucha is one of the most recognisable cultural symbols of Assam.** It is a white rectangular piece of cloth with embroidered red borders**.** During the Ahom King Syo-Kha-Pha's rule, this unique piece of cloth was worn by men as a head gear. In December 2022, gamosa received the GI tag. **So, pair 3 is correctly matched.**
- Recently, **six traditional crafts from Assam have bagged the GI tag**. These are Asharikandi terracotta craft, Pani Meteka craft, **Sarthebari metal craft,** Jaapi (bamboo headgear of rural Assam), Mishing handloom products and the Bihu dhol. **So, pair 4 is not correctly matched.**
    - The Sarthebari bell-metal industry is the second largest handicraft sector in Assam after bamboo craft.

So, only three of the pairs given above are correctly matched.
Therefore, option (c) is the correct answer.
Relevance: The Ashtalakshmi Mahotsav was recently held to celebrate the rich heritage of North East India.
Subject: Geography | Indian Social and Economic Geography
Tags: Factual

Q13.
Answer: d
Explanation:

- **Ancient genome sequencing** has revolutionized the ability to reconstruct expansions, migrations and admixture events in the ancient past and understand their impact on human genetic variation today. However, tracing history using genetic ancestry has remained challenging, particularly in historical periods for which the richest comparative information from history and archaeology often exists. This is

because ancestries in many geographical regions are often so similar as to be statistically indistinguishable with current approaches.

- **Twigstats** is a time-stratified **ancestry analysis tool** that greatly boosts the statistical power of existing methods and reduces statistical errors. It has determined more precisely than ever **individual-level ancestry at a very high resolution.** It is coded in **C++** and employs the **statistical language R**. It allows researchers to focus on a specific historical period with more fine-grained analysis than possible.

Therefore, option (d) is the correct answer.

Relevance: Recently, the scientists from the Francis Crick Institute in the UK and including multiple research groups across the UK, Japan and Sweden developed a new way to infer genome-wide genealogies.

Subject: Science & Technology | Biotechnology

Tags: Factual

Q14.

Answer: a

Explanation:

- **The United Nations General Assembly** established the **Internal Justice Council (IJC)** as part of the new internal justice system to help **ensure independence, professionalism, and accountability in the administration of the justice system of the United Nations**. **So, statement 1 is correct.**
- The IJC is composed of **five members** consisting of a staff representative, a management representative, and two distinguished external jurists, one nominated by the staff and one by management, and chaired by a distinguished jurist chosen by consensus by the four other members. The IJC members are appointed by the Secretary-General.
- **The United Nations Dispute Tribunal (UNDT), as the "first instance tribunal"** of the UN's internal justice system, is the tribunal UN system staff members apply to when they decide to challenge an administrative decision made by an entity over which the UNDT has Jurisdiction. **So, statement 2 is not correct.**

Therefore, option (a) is the correct answer.

Knowledge Box
• **The United Nations Appeals Tribunal (UNAT)** was established by the General Assembly in 2009, as part of the new UN internal justice system. • As the **second level appellate review tribunal** within the internal justice system, UNAT reviews appeals against judgments rendered by the **UN Dispute Tribunal**, the Dispute Tribunal of the **UN Relief and Works Agency for Palestine Refugees** in the Near East (UNRWA), as well as appeals of decisions taken by the Standing Committee acting on behalf of the **United Nations Joint Staff Pension Board** (UNJSPB).

Relevance: Recently the former Supreme Court Judge, Justice Madan B. Lokur, has been appointed Chairperson of the United Nations Internal Justice Council (IJC) for a term ending on November 12, 2028.

Subject: Polity | International Relations

Tags: Factual

Q15.
Answer: a
Explanation:

- The **Central Water Commission (CWC)** recently released a study titled **'Assessment of Water Resources of India 2024**'. The report estimated that India's average annual water availability, between 1985 and 2023, stood at **2,115.95** billion cubic meters (BCM).
- Taking the latest assessments into account, the **average annual per capita water availability for 2024 is 1,513 cubic meters** (for a projected population figure of 1.398 billion). This is still below the **1,700 cubic meters mark**.
- **Regional disparity:** According to the CWC report, **Brahmaputra (592.32 BCM), Ganga (581.75 BCM), and Godavari (129.17 BCM) are the top three basins** with the highest water availability across the country. Sabarmati (9.87 BCM), Pennar (10.42 BCM), and Mahi (13.03 BCM) were bottom three basins in terms of water availability.
    - The **annual groundwater recharge is significantly high in the Indus-Ganga-Brahmaputra alluvial** belt in the North, East and North East India covering the States of Punjab, Haryana, Uttar Pradesh, Bihar, West Bengal and valley areas of North Eastern States, where **rainfall is plenty and thick piles of unconsolidated alluvial formations are conducive for recharge. So, statements 1 and 2 are correct.**
    - The coastal alluvial belt, particularly the Eastern Coast, also has relatively high annual extractable groundwater resources.
      In western India, particularly Rajasthan and parts of northern Gujarat that have arid climate, the annual groundwater recharge is scanty.
    - Similarly, in major parts of the southern peninsula covered with **hard rock terrains** and parts of Central India, the annual groundwater recharge is low, primarily because of comparatively low infiltration and storage capacity of the rock formations prevailing in the region.

So, both Statement–I and Statement–II are correct and Statement–II explains Statement–I. Therefore, option (a) is the correct answer.

Knowledge Bank

Current vs. Previous assessments:
● According to the Ministry of Jal Shakti, the average annual per capita water availability for the year 2021 was 1,486 cubic meters based on CWC's 2019 study.
● The current water availability figures are **higher than previous assessments**. This is primarily due to methodological factors.
  ○ First, the new assessment takes into account Bhutan's contribution to the Brahmaputra, which was not included in the assessment conducted in 2019.
  ○ Second, while Nepal's contribution to the Ganga was only partially taken into account in the 2019 assessment, the current study includes it fully.
Falkenmark Indicator:
● According to the **Falkenmark Indicator or Water Stress Index**, a country would be deemed to be under **"water stress" if the per capita water availability in a**

country is below 1,700 cubic meters.

Relevance: The Central Water Commission (CWC) recently released a study titled 'Assessment of Water Resources of India 2024'.
Subject: Geography | Water Resources
Tags: Factual, Conceptual

Q16.
Answer: c
Explanation:

- **Jus soli**, Latin for "right of the soil," is the legal principle that a person's place of **birth** determines their citizenship. This is in contrast to **jus sanguinis**, or "right of blood," where citizenship is determined by the nationality of one or both parents. In jus soli countries, children born within the country's territory automatically become citizens, even if their parents are not citizens or legal residents. **So, statement 1 is correct.**
- The **US Constitution's 14th Amendment** guarantees birthright citizenship, stating that "all persons born or naturalized in the United States, and subject to the jurisdiction thereof, are citizens of the United States and of the state wherein they reside." India, on the other hand, **does not practice unconditional birthright citizenship**. A person born in India is considered a citizen only if at least one parent is a citizen, and the other is not an illegal migrant. These measures were implemented largely in response to illegal migration concerns. **So, statement 2 is correct.**

Therefore, option (c) is the correct answer.
Relevance: US President Donald Trump signed a list of executive orders including the order to end birthright citizenship.
Subject: Polity | International Relations
Tags: Factual

Q17.
Answer: b
Explanation:

- **INS Surat** is a next-generation stealth-guided missile destroyer under **Project 15B (Visakhapatnam-class)** and is equipped with **advanced artificial intelligence (AI)** capabilities to enhance decision-making, automation, and operational efficiency. It is designed for **multi-role operations** with a focus on network-centric warfare. **So, pair 1 is correctly matched.**
- **INS Nilgiri** is the **first of the seven stealth frigates** being built under the **Project 17A** program, which aims to enhance the Navy's capabilities with advanced stealth, automation, and **weapon systems.** The ship features cutting-edge indigenous technology, including an integrated **combat management** system. **So, pair 2 is correctly matched.**
- **INS Kalvari** is **not a nuclear-powered submarine** but a **diesel-electric attack submarine** of the Scorpène-class, built under **Project 75. India's first indigenous nuclear-powered submarine is INS Arihant.** INS Kalvari is designed for anti-surface and anti-submarine warfare capabilities. **So, pair 3 is not correctly matched.**
- **INS Vagsheer** is the sixth and **final submarine of the Kalvari-class** (Scorpène), which is being built under **Project 75.** It is equipped with advanced stealth features and **precision-guided weapon systems** to operate in both littoral and deep-sea environments.

So, only two of the pairs given above are correctly matched.
Therefore, option (b) is the correct answer.
Relevance: The Indian Navy is set to commission three advanced platforms — a Scorpene-class submarine (Vagsheer), a stealth destroyer (Surat), and a stealth frigate (Nilgiri) — on January 15, 2025, at the Naval Dockyard, Mumbai.
Subject: Science & Technology | Defence and Related Developments
Tags: Factual

Q18.
Answer: d
Explanation:

- The **spin glass** refers to **a magnetic state characterized by randomness.** It is a **disordered magnetic state.** It is a set of interacting magnetic moments, originating from spins, in which the magnetic interactions are randomly distributed. Thus, spin glasses are disordered magnets with random interactions that are, generally, in conflict with each other.
- It is formed when **magnetic elements**, like iron or manganese or chromium, **mix in tiny proportions** with non-magnetic elements such as **gold or copper or silver**.
- In the spin glass, the **'atomic spins' carried by magnetic elements freeze in random directions and locations,** between in the host lattice of nonmagnetic elements. They interact and influence each other's orientation in a 'strange' manner that leaves them frustrated due to conflicting commands for their orientation. This produces a very large number of patterns of **randomly oriented spins** that minimise the energy of the spin glass.

Therefore, option (d) is the correct answer.
Relevance: The Nobel Physics prize in 2024 was co-awarded to John J Hopfield, who gave a model for a neuronal network inspired by work done in the field of spin glass.
Subject: Science & Technology | Physics
Tags: Conceptual

Q19.
Answer: a
Explanation:

- The **Household Consumption Expenditure Survey** is designed to collect information on **consumption and expenditure of the households on goods and services**. The survey provides data required to assess trends in economic well-being. It enables us to determine and update the basket of consumer goods and services and weights used for the calculation of the **Consumer Price Index**.
- The average Monthly Per Capita Consumption Expenditure (MPCE) in rural and urban India in 2023-24 has been estimated to be Rs. 4,122 and Rs. 6,996, respectively without taking into account of the values of items received free of cost by the households through various social welfare programmes.
- Considering the imputed values of items received free of cost through various social welfare programmes, these estimates become Rs. 4,247 and Rs. 7,078 respectively, for rural and urban areas.
- The **urban-rural gap in MPCE has declined** to 71% in 2022-23 from 84% in 2011-12. It has **further come down to 70% in 2023-24** that confirms sustained momentum of consumption growth in rural areas.
- **Non-food items remain the major contributor to the household's average monthly expenditure** in 2023-24 with about 53% and 60% share in MPCE in rural and urban areas respectively.

Across all the States and UTs, the households are observed to spend more on non-food items. **So, statement 1 is not correct.**

- Conveyance, clothing, bedding & footwear, miscellaneous goods & entertainment and durable goods have a major expenditure share in non-food expenditure of the households in both rural and urban areas.
- Beverages, refreshments and processed food continue to have the major expenditure share in 2023-24 in the food items basket of the rural and urban households.

- **Consumption inequality**, both in rural and urban areas, has declined from the level of 2022-23. **The Gini coefficient has declined** to 0.237 in 2023-24 from 0.266 in 2022-23 for rural areas and to 0.284 in 2023-24 from 0.314 in 2022-23 for urban areas. **So, statement 2 is not correct.**
- **Among the States, MPCE is the highest in Sikkim** (Rural – Rs. 9,377 and Urban – Rs. 13,927) and it is the **lowest in Chhattisgarh** (Rural – Rs. 2,739 and Urban – Rs. 4,927). **So, statement 3 is correct.**
- **Among the UTs, MPCE is the highest in Chandigarh** (Rural – Rs. 8,857 and Urban – Rs. 13,425), whereas it is the l**owest in Dadra and Nagar Haveli and Daman and Diu** (Rs. 4,311) and Jammu and Kashmir (Rs. 6,327) in rural and urban areas, respectively.
- The rural-urban difference in average MPCE among the States is the highest in Meghalaya (104%) followed by Jharkhand (83%) and Chhattisgarh (80%).

So, only one of the statements given above is correct.
Therefore, option (a) is the correct answer.
Relevance: Household Consumption Expenditure Survey 2023-24 has been released recently.
Subject: Economics
Tags: Factual

Q20.
Answer: d
Explanation:

- The **Prevention Of Sexual Harassment at Workplace (POSH) Act 2013** aims to protect and prevent harassment of women at workplace and also redressal of the registered complaints. The POSH Act provides an **expansive definition of "workplace," covering:**
  - **Government institutions,** public sector units, and **autonomous bodies established**, owned, controlled, or funded directly or indirectly by the government. **So, option (a) is correct.**
  - **Private sector organisations,** hospitals, nursing homes, educational institutions, sports venues, and any other place visited by an employee during the course of employment.
  - **Domestic workplaces,** such as houses where women are employed as domestic help. **So, option (b) is correct.**
  - **Virtual workspaces,** extending protection to women working in online environments. **So, option (c) is correct.**
- **The POSH Act does not apply to political parties. So, option (d) is not correct.**
  - The applicability of the POSH Act to political parties is **not straightforward** due to the unique nature of their organisational structure.

- As per the **2022 judgment by the Kerala High Court** in Centre for Constitutional Rights Research and Advocacy vs State of Kerala & Ors., **political parties are not mandated** to establish an **Internal Complaints Committee** (ICC).
- Political parties **do not maintain a conventional employer-employee structure** with their members. Party workers are often volunteers or temporary recruits, which does not align with the definition of "employee" under the Act.
- Political parties **do not function as traditional businesses or establishments under the definition of "workplace" in the POSH Act.** Their operations are spread across various locations and often involve fieldwork, making it difficult to identify a specific "workplace."
- Political parties are governed by the Representation of the People Act, 1951, under which they are required to register with the Election Commission of India (ECI). The Act does not explicitly mandate compliance with the POSH Act.

Therefore, option (d) is the answer.
Relevance: The Supreme Court recently heard a PIL advocating for the application of The Sexual Harassment of Women at Workplace (Prevention, Prohibition and Redressal) Act, 2013 (commonly known as the POSH Act, 2013) to political parties.
Subject: Polity | Miscellaneous(pol)
Tags: Factual, Conceptual

Q21.
Answer: c
Explanation:

- The **Unified District Information System for Education Plus (UDISE+)** is an initiative of the **Ministry of Education**. It serves as an integrated online platform to **collect data related to schools across the country.** The platform aims to provide accurate and comprehensive data on school infrastructure, teachers, and students to facilitate better planning and policy implementation.
- Under UDISE+, a **unique 11-digit code** is assigned to all the recognized schools imparting formal education from **pre-primary to XII.** This code acts as a **unique identifier for schools** across India, covering both government and private institutions. **So, statement 1 is correct.**
- This unique identification number is **based on the geographical location of a school. So, statement 2 is correct.**
- **The code is permanent** in nature and once assigned to a specific school, it can't be changed. It **can only be modified by the central authorities if the school moves to a different district or State.** The code remains archived even if the school is permanently closed, ensuring data consistency and accuracy in tracking educational institutions. **So, statement 3 is correct.**

So, all three of the statements given above are correct.
Therefore, option (c) is the correct answer.
Relevance: The total enrolment of students has dropped by over a crore in 2023-24 as compared to the previous years, Unified District Information System for Education Plus (UDISE+) report released by the Ministry of Education (MoE) said recently.
Subject: Polity | Governance
Tags: Factual

Q22.
Answer: a

Explanation:

- The **United States of America (USA)** has acquired many territories since it began its journey as an independent nation after the Treaty of Paris in 1783.
- **The purchase of Louisiana was one of the largest acquisitions** of territory by the USA, which ended up doubling the country's area. Since 1762, Spain had owned the territory of Louisiana, which included 828,000 square miles. France acquired it from Spain in 1800 and took possession in 1802. Later **in 1803, the Louisiana Purchase was signed** between **France and the USA.** Under it, the USA purchased 828,000 square miles of land west of the Mississippi River for $15 million. **So, row 1 is correctly matched.**
- **Alaska Purchase:** In 1866, the Russian Government offered to sell the territory of Alaska to the United States. The two parties agreed on March 30, 1867, that **the US would pay Russia $7.2 million for the territory of Alaska.** The US thus got hold of nearly 600,000 square miles for less than **2 cents an acre**. **So, row 2 is not correctly matched.**
- **Texas** was originally part of Mexico. The Republic of Texas declared independence from Mexico on March 2, 1836. On March 1, 1845, the U.S. Congress passed the Joint Resolution for **the Annexation of Texas.** On June 23, 1845, the **Texas Congress approved a joint resolution (not Military annexation)** to annex Texas to the United States. On December 29, 1845, Texas was formally admitted to the USA as the 28th State. **So, row 3 is not correctly matched.**

So, in only one of the above rows is the given information correctly matched.
Therefore, option (a) is the correct answer.

Knowledge Box
The Gadsden Purchase: • It is a purchase of a roughly **30,000 square-mile region of present-day southern Arizona and southwestern New Mexico** that was **acquired by the United States**. • It was an agreement between the United States and Mexico, finalized in 1854. Under the treaty, **the United States agreed to pay Mexico $10 million for a 29,670 square mile portion of Mexico.** • It was the last major territorial acquisition in the contiguous United States.

Relevance: Donald Trump, the new President-elect of the United States, has expressed an interest in buying Greenland.
Subject: Polity | International Relations
Tags: Factual

Q23.
Answer: d
Explanation:

- The **Protected Area Regime (PAR)** in India, governed by the **Foreigners (Protected Areas) Order, 1958**, regulates the **entry of foreign nationals** into sensitive areas near international borders. **So, statement 1 is correct.**
- It applies to regions **between the "Inner Line" and international borders**, covering States like Arunachal Pradesh, Manipur, Mizoram, Nagaland, Sikkim, and others. Foreign nationals, **except**

Bhutanese, require a Protected Area Permit (PAP) issued by the Ministry of Home Affairs, **valid for 10 days. So, statement 2 is correct.**

- The **Inner Line Permit (ILP)** is an internal travel document **required by Indian citizens** to enter certain northeastern states **to protect indigenous communities,** whereas the PAR applies solely to foreign nationals, regulating their access to sensitive border regions for security reasons. **So, statement 3 is correct.**

Therefore, option (d) is the correct answer.

Relevance: Recently, the Union Government reinstated the Protected Area Regime (PAR) in three Northeastern States - Manipur, Mizoram, and Nagaland.

Subject: Polity | Scheduled and Tribal Areas

Tags: Factual

Q24.

Answer: a

Explanation:

- **Artesian conditions** occur when **groundwater** is **confined under pressure** within an aquifer that is sandwiched between two **impermeable layers of rock** or clay. Under these conditions, water in the aquifer is forced to rise above the level of the aquifer when a well is drilled into it due to the **hydraulic pressure** exerted by water entering the aquifer from a recharge area at a higher elevation. **So, point 1 is correct.**
- An artesian aquifer requires a **recharge zone at a higher elevation** to create a pressure gradient. Water infiltrates at the recharge area and flows downward due to gravity, **increasing the pressure in the confined aquifer.** This pressure is what allows water to rise above the land surface when tapped. **So, point 2 is correct.**
- **High porosity** means that a rock layer has **more spaces to store water,** but artesian conditions require impermeable layers to maintain pressure. A highly porous layer would allow water to escape, **preventing the formation of artesian conditions. So, point 3 is not correct.**
- Artesian aquifers must remain confined under pressure. If the aquifer is directly **exposed to the atmosphere,** the **pressure would dissipate**, and the water would not rise naturally. Unconfined aquifers **do not create artesian conditions. So, point 4 is not correct.**

Therefore, option (a) is the correct answer.

Relevance: Recently, residents of Taranagar village in Jaisalmer, Rajasthan, witnessed underground water gush out unexpectedly. Geologists from Rajasthan's State Ground Water Department attributed this phenomenon to an "artesian condition."

Subject: Geography | Water Resources

Tags: Factual

Q25.

Answer: b

Explanation:

- **Aerogels** are a class of **solid materials** that are **extremely lightweight and porous.** They are **synthetic materials**, **not naturally occurring.** They are produced through a process involving the removal of liquid from a gel under **supercritical drying** conditions, leaving behind a porous structure. **So, statement 1 is not correct.**

- They have an extremely low density and are **composed of up to 99% air** by volume, making them one of the **lightest solid materials** known. This characteristic gives them exceptional **thermal** and **acoustic insulation** properties. **So, statement 2 is correct.**
- They are extensively used in **space exploration missions,** including those by NASA, for their **lightweight, thermal insulation,** and **high impact absorption capabilities.** They have been used in spacecraft insulation and for capturing space dust particles in missions such as **NASA's Stardust probe. So, statement 3 is correct.**

Therefore, option (b) is the correct answer.

Relevance: A group of researchers from Pune have developed a novel hybrid aerogel capable of easily extracting gold from electronic waste.

Subject: Science & Technology | Space and Related Developments

Tags: Factual

Q26.

Answer: b

Explanation:

- The **Egyptian cotton leafworm (*Spodoptera littoralis*)** is a highly **polyphagous** moth species, feeding on over 40 plant families, including economically significant crops such as cotton, tomatoes, maize, and peppers. **Native to Africa,** it has spread to the **Middle East** and parts of **Mediterranean Europe.**
- Recent studies have shown that they **can hear ultrasonic sounds emitted by stressed plants**, such as tomato plants, and use these acoustic cues **to avoid laying eggs** on them. This ability enhances their **reproductive success** by selecting healthier plants that provide optimal food resources for their offspring.
- The study highlights the **intricate interactions** between insects and plants, suggesting that acoustic emissions play a crucial role in **ecological decision-making.**

Therefore, option (b) is the correct answer.

Relevance: A recent study confirmed that a moth species, the Egyptian cotton leafworm, can hear sounds emitted by stressed plants.

Subject: Environment | Plant and Animal Anatomy

Tags: Factual

Q27.

Answer: c

Explanation:

- A **Joint Parliamentary Committee (JPC)** is an **ad-hoc** committee set up by the Parliament for a **special purpose,** like the **detailed scrutiny of a subject or Bill**. As the name suggests, it has **members from both the Houses** and from the ruling parties and the opposition. It is dissolved after its term ends or its task has been completed.
- The **proceedings and findings** of the JPC are **confidential,** except in cases of public interest.
- JPC recommendations are **only advisory in nature and not legally binding** on the government. The government has the discretion to accept, modify, or reject the recommendations. **So, statement 2 is not correct.**
- However, the **Government must submit an Action Taken Report (ATR)** to Parliament detailing the steps taken based on the JPC's recommendations. **So, statement 1 is correct.**

So, Statement–I is correct, but Statement–II is incorrect.
Therefore, option (c) is the correct answer.
Relevance: A 31-member Joint Parliamentary Committee (JPC) has been formed to review the Bills related to the One Nation One Election.
Subject: Polity | Parliament
Tags: Factual, Conceptual

Q28.
Answer: a
Explanation:
Key findings of India State of Forest Report 2023:

- The **Forest and Tree cover** of the country is 8,27,357 sq km which is **25.17** percent of the geographical area of the country, consisting of 7,15,343 sq km (21.76%) as forest cover and 1,12,014 sq km (3.41%) as tree cover. **So, option (b) is correct.**
- **19 States/UTs have above 33 percent** of the geographical area under forest cover. Out of these, **8 States/UTs** namely Mizoram, Lakshadweep, A & N Island, Arunachal Pradesh, Nagaland, Meghalaya, Tripura, and Manipur have forest cover **above 75 percent**. The **National Forest Policy, 1988** envisages that **33% of the geographical area should be under forest or tree cover. So, option (a) is not correct.**
- Area wise top three States having largest forest and tree cover are **Madhya Pradesh** (85,724 sq km) followed by Arunachal Pradesh (67,083 sq km) and Maharashtra (65,383 sq km). Area wise top three states having largest forest cover are **Madhya Pradesh** (77,073 sq km) followed by Arunachal Pradesh (65,882 sq km) and Chhattisgarh (55,812 sq km).
- In terms of percentage of forest cover with respect to total geographical area, **Lakshadweep (91.33 percent) has the highest forest** cover followed by Mizoram (85.34 percent) and Andaman & Nicobar Island (81.62 percent). **So, option (c) is correct.**
- Regarding status on achievement of target under NDC related to carbon sequestration, the current assessment shows that **India's carbon stock has reached 30.43 billion tonnes of CO_2 equivalent**; which indicates that as compared to the base year of 2005, India has already reached 2.29 billion tonnes of additional carbon sink as against the target of 2.5 to 3.0 billion tonnes by 2030. **So, option (d) is correct.**

Therefore, option (a) is the answer.
Relevance: Minister for Environment, Forest and Climate Change, Shri Bhupender Yadav released the 'India State of Forest Report 2023 (ISFR 2023) today at Forest Research Institute, Dehradun.
Subject: Geography | Natural Vegetation and Wildlife
Tags: Factual

Q29.
Answer: c
Explanation:

- ***Helicobacter pylori* is a bacterium that affects over 43% of the world's population** with a wide range of **gastrointestinal disorders**, including peptic ulcers, gastritis, dyspepsia and even gastric cancer. **So, statement 1 is correct.**
- Recently, scientists have found a new method for detection of ***H. pylori*** and its mutations in dyspeptic patients. The technique focuses on a method called **CRISPR-based diagnostics (CRISPRDx)**. This

technique utilizes the CRISPR-Cas9 system, known for its ability to recognize and cleave specific DNA sequences, to **detect the presence of *H. pylori*** and identify mutations associated with antibiotic resistance. **So, statement 2 is correct.**

- The researchers used an engineered Cas9 protein, en31-FnCas9, to detect H. pylori and its mutations in gastric biopsy samples. This approach, combined with lateral flow-based test strip assays (**Feluda Test**), offers a rapid visual readout of the infection status and antibiotic resistance profile of *H. pylori*.
- **Resistance to clarithromycin is a major concern in treating *H. pylori* infections**. This resistance is often caused by point mutations in the 23S ribosomal RNA coding gene of the bacterium. This resistance necessitates repeated diagnostic tests and multiple courses of different antibiotic combinations for eradication, posing a global public health threat.

Therefore, option (c) is the correct answer.
Relevance: Scientists have found a new method for detection of *H. pylori* and its mutations in dyspeptic patients.
Subject: Science & Technology | Health and Diseases
Tags: Factual, Conceptual

Q30.
Answer: a
Explanation:

- **Donyi Polo,** meaning **"Sun and Moon,"** is an **indigenous faith** followed by various tribes in **Arunachal Pradesh**, particularly **the Tani group of tribes.**
- The faith is rooted in **nature worship,** emphasizing the reverence of natural elements such as mountains, rivers, and animals.
- It promotes **environmental stewardship,** moral values, and cultural identity through rituals, festivals, and traditional practices.
- Donyi Polo plays a significant role in preserving the **unique heritage and traditions** of the indigenous communities in the region.

Therefore, option (a) is the correct answer.
Relevance: The Arunachal Pradesh Government is reviving the 1978 Freedom of Religion Act, sparking debates on religious freedom, indigenous culture and State's involvement in faith matters.
Subject: Geography | Indian Social and Economic Geography
Tags: Factual

Q31.
Answer: d
Explanation:

- A **tactical sonar system for transmitting submarine activity**, the sonobuoy is a core technology for anti-submarine warfare**. Sonobuoys are expendable, electro-mechanical acoustic sensors that relay underwater sounds emitted from ships and submarines**. They remain active for about 24 hours and help in the detection, classification, and prosecution of adversarial ships and submarines. A naval helicopter or fixed-wing aircraft generally drops sonobuoys in a pattern
- Sonobuoys have an inflatable system with a radio transmitter that remains on the surface for communication with the ship or aircraft tracking it while sensors descend below the surface to a predetermined depth. It then relays acoustic information back to those monitoring them. **A group of sonobuoys deployed in a pattern can find out the exact location of the submarine which then can be tracked by other systems. So, statement 1 is correct.**

- Some sonobuoys are designed to be deployed in passive mode and some in active mode. **Active sonobuoys emit sound energy and receive the echo, based on which it transmits information** back to the aircraft. Passive sonobuoys, on the other hand, only listen for sounds coming from ships or submarines. They then transmit the sound back to the aircraft. **So, statement 2 is correct.**
- For special-purpose sonobuoys, the detections are of magnetic anomalies and electrical fields. **In addition, they measure various environmental parameters such as water temperature along the depth, barometric pressure, and height of ocean waves**. Generally, sonobuoys are deployed from aircraft equipped with launchers for their deployment. In addition, these aircraft have electronic equipment for the reception and subsequent processing of data transmitted by the sonobuoys. **So, statement 3 is correct.**

Therefore, option (d) is the correct answer.

Relevance: The US announced that the State Department approved the sale of sonobuoys–a critical expendable system for anti-submarine warfare–for an estimated cost of $52.8 million to the Indian Navy which will be deployable from MH-60R helicopters.

Subject: Science & Technology | Defence and Related Developments

Tags: Factual, Conceptual

Q32.

Answer: b

Explanation:

- **Tipeshwar Sanctuary** is located in the Yavatmal district of Maharashtra state in an area known as Pandarkawda. It is named after the goddess Tipai whose temple is located inside the forest. **The four rivers flow from this wildlife sanctuary – Bhima, Purna, Krishna, and Tapti**. Hence, it is also known as the Green Oasis of Eastern Maharashtra. **So, pair 1 is correctly matched.**
- The **Bhindawas Wildlife Sanctuary** is situated in Jhajjar district, Haryana. It was officially designated as a bird sanctuary by the Government of India on June 3, 2009. The sanctuary, recognized as a Ramsar site, plays a significant role in global efforts to conserve wetlands. **The sanctuary is an essential part of the ecological corridor that runs along the Sahibi River route.** This corridor extends from the Aravalli Hills in Rajasthan to the Yamuna River, covering several vital natural sites. **So, pair 2 is not correctly matched.**
- Two rivers **Chinnar and Pambar** run through the **Chinnar Wildlife Sanctuary which supports the ecology of the sanctuary**. The Pambar, which originates in the Anamudi hills, traverses the Eravikulam National Park and flows down into the Chinnar sanctuary. **So, pair 3 is correctly matched.**

So, only two of the pairs given above are correctly matched.

Therefore, option (b) is the correct answer.

Relevance: Recently a male tiger born in the Tipeshwar Wildlife Sanctuary in Yavatmal has traveled an unusual 500 kilometers, reaching the Solapur district.

Subject: Environment | Biodiversity and Conservation

Tags: Factual

Q33.

Answer: c

Explanation:

- **South Pole-Aitken Basin (SPA)** is an **impact crater** on the **far side of the Moon.** Measuring some 2,500km (1,600 miles) in diameter, and up to 8.2km deep in some places, it's the **oldest and deepest basin on the Moon,** and one of the **largest known impact craters** in the entire Solar System. **So, statement 1 is correct**
- The South Pole-Aitken Basin is about **4 billion years old.** The SPA Basin is significant for planetary scientists because its great depth may have exposed materials from the Moon's lower crust or even upper mantle, which are otherwise buried beneath the surface. Studying these materials can provide **insights into the Moon's internal structure, its early geological history, and the impact events that shaped its evolution.** China's Chang'e-4 mission has been studying this basin for these reasons**. So, statement 2 is correct**

Therefore, option (c) is the correct answer.
Relevance: Scientists detected a colossal mass of material beneath the Moon's South Pole-Aitken basin
Subject: Science & Technology | Space and Related Developments
Tags: Factual

Q34.
Answer: b
Explanation:

- **Automated & Intelligent Machine-aided Construction (AIMC)** is designed to enhance construction efficiency by using **automated & intelligent machines equipped with GPS, sensors, and real-time data transmission.** The system helps improve construction speed, quality control, and durability of roads. It is currently being piloted in the Lucknow-Kanpur Expressway project by the National Highways Authority of India (NHAI). **So, statement 1 is correct.**
- AIMC **does not rely on Artificial Intelligence** (AI) for independent decision-making. Instead, it involves GPS-based automation, sensor technology, and real-time data transmission to improve construction efficiency. S**o, statement 2 is not correct.**
- AIMC i**ntegrates automated surveying and monitoring systems,** which reduces the need for manual inspections at every stage. The system ensures **continuous real-time tracking** of the construction process, allowing stakeholders like the Ministry of Road Transport & Highways (MoRTH), NHAI, to **monitor project** progress efficiently. This helps in minimizing delays and ensuring quality control throughout the construction process. **So, statement 3 is correct.**

Therefore, option (b) is the correct answer.
Relevance: NHAI is implementing AIMC on a pilot basis in the 63-km long under-construction Lucknow-Kanpur Expressway Project
Subject: Economics | Infrastructure
Tags: Factual, Analytical

Q35.
Answer: d
Explanation:

- **Coconut oil** is widely used for **both cooking and personal care in India.** Earlier, its **taxation was based on its dual usage but now**, it is based on the legal classification, defined by the recent Supreme Court decision.

- The **Supreme Court** ruled that **coconut oil is an edible oil** irrespective of its packaging size. Earlier, the Central Board of Excise and Customs (CBEC) had classified coconut oil sold in **small containers (≤200 ml) as hair oil for taxation**, but this was overturned by the Court and later withdrawn in 2015.
- The Supreme Court **rejected the 'common parlance test'** and clarified that **coconut oil should not be classified based on packaging size** but should be taxed under the edible oil category. Edible oils currently attract a lower rate of Goods and Services Tax (5%) than hair care products (18%).

Therefore, option (d) is the correct answer.
Relevance: SC verdict on coconut oil-coconut oil is an edible oil, and should be taxed at a lower rate.
Subject: Economics | Taxation
Tags: Factual, Analytical

Q36.
Answer: c
Explanation:

- **Captagon** was first developed in 1961 as a **psychoactive medicine** containing fenethylline. It was prescribed for conditions such as **narcolepsy, fatigue** and **Attention Deficit Hyperactivity Disorder (ADHD).** It functioned as a central nervous system stimulant but was later classified as a controlled substance due to its potential for abuse. **So, statement 1 is correct.**
- The illegal, modern version of Captagon is often mixed with highly addictive stimulants and other harmful substances. These compounds can cause **long-term damage to brain function, particularly affecting impulse control, judgment** and **rational thinking.** Captagon has been **used by militant groups** like ISIS, Hezbollah, etc. to enhance aggression and endurance, further demonstrating its neurotoxic effects. **So, statement 2 is correct.**
- **The Captagon Act** was introduced by the **United States** in 2022. It focuses on targeting illicit production and trafficking of the drug.
- After the fall of the Bashar al-Assad regime in Syria, large stockpiles of the illicit drug captagon have reportedly been uncovered.

Therefore, option (c) is the correct answer.
Relevance: The stockpiles of Captagon, found by Syrian rebels, are believed to be linked to al-Assad military headquarters, implicating the fallen regime in the drug's manufacture and distribution.
Subject: Science & Technology | Recent Developments
Tags: Factual

Q37.
Answer: c
Explanation:

- **The Panama Canal,** completed in **1914,** is a 50-mile sea-level passage connecting the **Atlantic and Pacific Oceans** through the Isthmus of Panama.
- Initially attempted by the French in the 1880s, the project failed due to disease and engineering challenges. The **United States** took over in 1904, following Panama's independence from Colombia, and **completed the canal,** revolutionizing global maritime trade by significantly reducing travel time for ships**.**
- Strategically and economically crucial, the canal enhances **global trade efficiency,** especially for the U.S., China, and Latin American countries**.**

- **In 1999,** control of the canal was **transferred from the U.S. to Panama,** which continues to manage and expand its operations today. The treaties associated with the canal are:
- **Clayton-Bulwer Treaty (1850)** was signed between the **U.S. and Britain** to prevent either country from having exclusive control over a canal in Central America. It was later superseded by the Hay-Pauncefote Treaty**.**
- **The Hay–Pauncefote Treaty** was a 1901 agreement between the **United States and Great Britain.** The Treaty licensed the United States to **build and manage its own canal.** The treaty cancelled the Clayton–Bulwer Treaty of 1850, which had previously prohibited either country from building a canal without the other's consent.
- **Hay-Bunau-Varilla Treaty (1903)** was signed between the newly independent Republic of **Panama and the U.S.,** granting the U.S. a **10-mile-wide canal zone in exchange** for financial compensation and a U.S. guarantee of Panamanian independence**.**
- **The Hay-Herrán Treaty (1903)** was signed between the **U.S. and Colombia** to allow the **U.S. to build the canal.** However, it was **rejected by the Colombian Senate,** leading to U.S. support for Panamanian independence.

Therefore, option (c) is the correct answer.
Relevance: Donal Trump's recent statement on Panama Canal.
Subject: Polity | International Relations
Tags: Factual

Q38.
Answer: b
Explanation:

- **The United Nations Disengagement Observer Force (UNDOF) does not monitor the ceasefire between Armenia and Azerbaijan.** Instead, it is tasked with monitoring the ceasefire between **Israel and Syria** in the Golan Heights. The conflict between Armenia and Azerbaijan, particularly over the Nagorno-Karabakh region, is not under the mandate of UNDOF. **So, statement 1 is not correct.**
- UNDOF is **not located in the Nagorno-Karabakh region**. It is stationed in the **Golan Heights,** a region that is disputed between Israel and Syria. The Nagorno-Karabakh region is a separate area that has been the focal point of conflict between Armenia and Azerbaijan. **So, statement 2 is not correct.**
- **UNDOF's** main responsibility is to **monitor and maintain the ceasefire** between Israel and Syria in the Golan Heights. It was established in **1974 to supervise the disengagement of forces following the 1973 Yom Kippur War,** ensuring that both parties adhere to the terms of the ceasefire and disengagement agreement**.**
- **India has been a significant contributor to UNDOF** since its establishment in 1974. India is one of the largest contributors to UN peacekeeping operations and has provided troops to UNDOF for several decades. Indian personnel have played a key role in maintaining peace and monitoring the ceasefire in the Golan Heights**. So, statement 3 is correct.**

Therefore, option (b) is the correct answer.
Relevance: UN Secretary General pays tribute to Brigadier Amitabh Jha who led UN peacekeeping force
Subject: Polity | International Relations
Tags: Factual

Q39.
Answer: c
Explanation:

- The **Chancay port, located in Peru,** has recently gained significant attention due to its **strategic partnership with China**, part of the Belt and Road Initiative (**BRI**).
- Chinese President Xi Jinping **inaugurated the port**, emphasizing its role as the starting point for a **"new land-sea corridor" between China and Latin America**. This $3.6 billion project, backed by Chinese investment, is set to transform **Chancay into a key trade hub,** enhancing the **flow of goods from South America to Asia**, particularly benefiting Peru and its neighbors.
- Chancay's location is crucial, as it **reduces the transportation time for goods between Latin America and Asia**, which previously required transshipment through the U.S. or Mexico.
- The port's development will help **reduce logistics costs, generate jobs, and boost Peru's economy.** It will handle key exports like copper, soybeans, and lithium, the latter being in high demand for electric vehicles, further strengthening its role in global trade.
- This partnership also raises **geopolitical concerns**, particularly in the U.S., where officials worry about **China's growing influence in the region**.
- The port's capabilities to handle some of the **largest vessels in the world (up to 18,000 TEUs)** solidify Chancay's strategic importance. Additionally, the port's potential connections to Brazil are expected to have broader implications for regional trade and global shipping, making it a key player in China's growing presence in South America.

Therefore, option (c) is the correct answer.
Relevance: Chinese President Xi Jinping inaugurated Peru's Chancay Port.
Subject: Polity | International Relations
Tags: Factual, Analytical

Q40.
Answer: b
Explanation:

- **Gas flaring** is not the process of capturing and storing excess natural gas. Instead, it refers to the **burning of the natural gas that is released during oil extraction.** This practice is often adopted due to economic, technical, safety, and regulatory constraints. Gas flaring results in wastage of a valuable energy resource and contributes to air pollution and climate change. **So, statement 1 is not correct.**
- **Oil extraction involves high pressures**, and sudden or unpredictable pressure surges can cause industrial accidents, including explosions. **Flaring is used as a pressure-release mechanism to safely dispose of excess gas** and prevent hazardous situations. Without flaring, gas buildup could lead to fires, equipment failure, and operational risks. S**o, statement 2 is correct.**
- **Methane (CH_4) is a major greenhouse gas released when gas flaring is incomplete or inefficient.** Methane is over 80 times more potent than CO_2 in causing global warming over a 20-year period. Gas flaring combustion efficiency is usually around 98%, but even a small inefficiency leads to methane leakage, worsening climate change. In 2023, gas flaring contributed to over 350 million tons of CO_2-equivalent emissions, including methane and black carbon. **So, statement 3 is correct.**

Therefore, option (b) is the correct answer.

Knowledge Box
Zero Routine Flaring by 2030 Initiative: ● Launched in 2015 by the World Bank and UN Secretary General, the ZRF Initiative commits governments and oil companies to end routine flaring no later than 2030. ● It encourages governments and oil companies to stop flaring by capturing and utilizing gas for productive uses such as power generation. Increasing flaring would not reduce reliance on coal and diesel; instead, using captured gas as an alternative fuel would achieve that goal. **Global Flaring and Methane Reduction Partnership (GFMR):** ● It is an initiative of the World Bank. ● It was formerly known as the Global Gas Flaring Reduction Partnership (GGFR). ● It is a multi-donor trust fund composed of governments, oil companies, and multilateral organizations committed to ending routine gas at oil production sites across the world and reducing methane emissions from the oil and gas sector to near zero by 2030.

Relevance: The World Bank has launched the Global Flaring & Methane Reduction (GFMR) Partnership.
Subject: Environment | Climate Change
Tags: Factual, Analytical

Q41.
Answer: c
Explanation:

- **Boustrophedon** is an ancient writing style where **alternate lines are written in opposite directions:** one line from left to right, the next from right to left. The term derives from the **Greek words "bous" (ox) and "strophē" (turn)**, likening the text's direction to an ox's path while plowing a field.
- **Indus Script:** The Indus Valley Civilization utilized the **boustrophedon style** in its writing system, as evidenced by inscriptions found on seals, pottery, and other artifacts excavated from sites such as **Harappa and Mohenjo-Daro.** The script, composed of pictographic symbols and geometric patterns, remains undeciphered so far. **So, point 1 is correct.**
- **Greek Script:** Early Greek inscriptions, particularly from the 6th century BCE, frequently employed the **boustrophedon style.** This writing method gradually fell out of use as the Greek script **evolved** towards a consistent left-to-right orientation. **So, point 2 is correct.**
- **Hittite Script:** The Hittite civilization (circa 1600–1178 BCE), centered in **Anatolia** (modern-day Turkey), used **boustrophedon writing** in their cuneiform inscriptions, particularly on legal and administrative documents. **So, point 3 is correct.**
- **Kadamba script:** It was derived from the Brahmi script and generally written in a **consistent direction (usually left-to-right). So, point 4 is not correct.**

- **Luwian Hieroglyphic Script:** The Luwian civilization (circa 2000–700 BCE), contemporaries of the Hittites, also employed the **boustrophedon style** in their hieroglyphic inscriptions found on stone monuments and metal tablets. **So, point 5 is correct.**

Point To Remember
The Kharosthi script was traditionally written from right to left, akin to many ancient scripts. However, **some inscriptions have been found** with a **boustrophedon style**, which changes direction from line to line.

So, only four of the above were written in the Boustrophedon style.
Therefore, option (c) is the correct answer.
Relevance: Tamil Nadu's Chief Minister recently announced a $1 million prize for successfully deciphering the Indus Valley script.
Subject: History | Indus Valley Civilization
Tags: Factual

Q42.
Answer: c
Explanation:

- As per the **India State of Forest Report (ISFR) 2023**, the top four states that recorded the **maximum increase in forest and tree cover** are **Chhattisgarh** (684 sq km), **Uttar Pradesh** (559 sq km), **Odisha** (559 sq km), and **Rajasthan** (394 sq km).
    - **Chhattisgarh witnessed the highest increase**, followed closely by Uttar Pradesh and Odisha, which saw equal growth, while Rajasthan experienced a comparatively lower increase.
- The report highlights the efforts of these states in **afforestation and sustainable forest management,** contributing to the **overall increase of 1,445 sq km** in the country's forest and tree cover compared to the 2021 assessment.

So, the correct sequence would be Chhattisgarh - Uttar Pradesh - Rajasthan.
Therefore, option (c) is the correct answer.
Relevance: Recently, the Minister for Environment, Forest and Climate Change released the 'India State of Forest Report 2023 (ISFR 2023) at Forest Research Institute, Dehradun.
Subject: Geography | Natural Vegetation and Wildlife
Tags: Factual

Q43.
Answer: b
Explanation:

- **Opinion trading platforms** are applications that solicit people's opinions on various topics, typically in a **'Yes' or 'No' format. Users can bet money** based on their answers and potentially win.
- Platforms like **Probo** and **MPL Opinion Trading** have seen significant growth, attracting funding of over ₹5,000 crore. Additionally, the **user base is expanding,** with Probo alone boasting more than 5 crore users.
- In the **Indian legal landscape,** opinion trading is **not generally considered a game of skill** and is often viewed as **more akin to gambling** due to the significant element of chance involved, **raising concerns about its legality** under existing gambling laws. However, the exact classification can be

debated depending on the specific platform and its design, with some arguing that certain aspects might involve skill if properly structured. **So, statement 1 is not correct.**

- Currently, there is **no dedicated regulatory body in India** to oversee opinion trading, leading to legal ambiguities and regulatory challenges. While the **IT Ministry covers certain aspects,** a comprehensive framework is lacking. **So, statement 2 is correct.**

Therefore, option (b) is the correct answer.
Relevance: The government is closely scrutinizing opinion trading platforms, questioning their legitimacy based on the criteria of 'game of skill' and 'game of chance.'
Subject: Economics | Miscellaneous(eco)
Tags: Factual

Q44.
Answer: c
Explanation:

- **Tinnitus** is the **perception of sound without an external source,** meaning only the affected individual hears it. It is commonly caused by **age-related hearing loss,** ear injuries, or circulatory issues.
- **Symptoms include phantom noises** such as buzzing, roaring, or clicking, along with sleep disturbances, anxiety, depression, and social difficulties.
- It can significantly impact **mental well-being** and hence, requires a tailored approach for relief.
- **Treatment options include** hearing aids, sound-masking devices, medications, and coping strategies to manage the condition.

Therefore, option (c) is the correct answer.
Relevance: Researchers at the Indian Institute of Technology (IIT) Bombay have developed an affordable device to diagnose and manage tinnitus.
Subject: Science & Technology | Health and Diseases
Tags: Conceptual

Q45.
Answer: d
Explanation:

- The **Annual Survey of Unincorporated Sector Enterprises (ASUSE) 2022-2023** is a survey conducted by the **Ministry of Statistics and Programme Implementation** to gather data on the economic and operational characteristics of **non-agricultural, unincorporated businesses in India,** including those in manufacturing, trade, and other services sectors, providing insights into the performance of small-scale, non-registered businesses across the country.
- **Some key findings of the ASUSE 2022-2023:**
    - Number of estimated establishments belonging to unincorporated 'manufacturing', 'trade' and 'other services sector' **grew from 5.97 crore during April, 2021-March, 2022 to 6.50 crore** during the survey period of ASUSE 2022-23, which is, October, 2022-September, 2023. Out of these estimated 6.50 crore establishments, **about 3.56 crore belong to rural areas** and about 2.94 crore pertain to urban areas. **So, statement 1 is correct.**
    - At all-India level, **Uttar Pradesh (13.82%), West Bengal (12.03%)** and **Maharashtra (9.37%)** have produced the most number of unincorporated **non-agricultural establishments. So, statement 2 is correct.**

- There are as many as 3 major States such as, **Telangana (43.32%), West Bengal (32.74%), and Karnataka (31.42%)** where more than 30% of the proprietary establishments have been **run by women entrepreneurs. So, statement 3 is correct.**

Therefore, option (d) is the correct answer.

Relevance: Recently, the Ministry of Statistics and Programme Implementation has released the results of Annual Survey of Unincorporated Sector Enterprises (ASUSE) for 2022-23.

Subject: Economics | Industry

Tags: Factual

Q46.

Answer: b

Explanation:

- **Community Sourced Impact-Based Flood Forecast and Early Warning System (CoS-IT-FloWS)** is an open-source Decision Support System developed by EQUINOCT under UNICEF's **Innovation Venture Capital Fund.**
- It provides **river basin-level impact-based flood forecasting** and multi-hazard warnings using real-time meteorological and infrastructural data, such as rainfall, groundwater, and reservoir levels. Leveraging **hydrological models** and machine learning, it offers **hyper-local forecasts.**
- Key features include a **community weather data collection app ("Gather"),** an AI-powered real-time monitoring dashboard and an open-source back-end.
- The project promotes transparency and collaboration through **GitHub under the MIT license,** encouraging global contributions and improvements.

Therefore, option (b) is the correct answer.

Relevance: Recently, Kerala launched a new system called Cos-it-FloWs, designed to collect hyper-local data for flood forecasting in the flood-prone Periyar and Chalakudi river basins.

Subject: Geography | Miscellaneous(geo)

Tags: Factual

Q47.

Answer: b

Explanation:

- **India is the second largest tobacco producer behind China.** The country has around 0.45 million hectares of area under tobacco cultivation. About **10% of the total area** under tobacco cultivation is in India. Globally, it accounts for **9% of the total tobacco production. So, statement 1 is not correct.**
- **India is the 2nd largest exporter of unmanufactured tobacco (quantity terms) after Brazil.** It exports various types of tobacco and tobacco products such as stripped, wholly stemmed, cigar cheroots, smoking tobacco, homogenized, flue-cured, sun-cured, extract and essence, FCV tobacco, unmanufactured tobacco and various tobacco products. **So, statement 2 is correct.**
- India produces various types of tobacco: **Flue-cured Virginia (FCV) tobacco,** Country tobacco, Burley tobacco, Bidi tobacco, Rustica tobacco, Hookah, Cigar rapped, Cheroot, Burley, Oriental, Chewing tobacco, etc. **India is the 4th largest producer of FCV (Flue-Cured Virginia) tobacco** after China, Brazil, and Zimbabwe. **So, statement 3 is correct.**

Therefore, option (b) is the correct answer.

Relevance: During 2023-24 the value of Indian tobacco exports reached Rs.12005.89 crore (1449.54 in US dollars).
Subject: Economics | Agriculture and Food Management
Tags: Factual

Q48.
Answer: c
Explanation:

- **Precision Formation Flying (PFF)** is a highly **coordinated maneuver where multiple satellites operate in a controlled configuration** to achieve a fixed relative position in space.
- **The Proba-3 mission,** launched by Indian Space Research Organisation (ISRO) for the European Space Agency (ESA), will be the **first-ever attempt at this technique.**
- The **Proba-3 mission** is designed with **two satellites** that will be launched together, separate from each other and then **fly in tandem**. They will then form a solar coronagraph, an instrument that helps block out the bright light emitted by the Sun to reveal the objects and atmosphere around it. It will help to **observe the Sun's corona,** the outermost and hottest part of the Sun's atmosphere.

Therefore, option (c) is the correct answer.

Relevance: ISRO to launch Proba-3, the advanced European solar mission that uses Precision Formation Flying (PFF) technique for the first time.
Subject: Science & Technology | Space and Related Developments
Tags: Factual, Analytical

Q49.
Answer: c
Explanation:

- **Wildlife Crime Control Bureau (WCCB)** is a **statutory multi-disciplinary body** established by the Government of India under the Ministry of Environment and Forests, to combat organized wildlife crime in the country. The Bureau has its headquarter in New Delhi and five regional offices at Delhi, Kolkata, Mumbai, Chennai and Bhopal; three sub-regional offices at Guwahati, Amritsar and Cochin; and five border units at Ramanathapuram, Gorakhpur, Motihari, Nathula and Moreh. S**o, statement 1 is correct.**
- The **WCCB** was established in 2007 under the provisions of **the Wildlife (Protection) Act, 1972.** The Wildlife (Protection) Act, 1972 provides the legal framework for the establishment and functioning of the Bureau to **combat organized wildlife crimes** in India.
- It primarily functions as an **intelligence and coordination agency** and **does not have prosecutorial powers.** It collects intelligence, disseminates it to relevant enforcement authorities such as **forest departments, police, and customs**, and assists in the investigation and prosecution process. The actual prosecution of offenders is **carried out by state wildlife authorities** and law enforcement agencies under the provisions of the Wildlife (Protection) Act, 1972. **So, statement 2 is not correct.**
- It is the designated **nodal agency** for the enforcement of the **Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES)** in India. It assists customs and enforcement agencies in the regulation of international trade in wildlife parts and products as per CITES regulations and **India's EXIM policy. So, statement 3 is correct.**

Therefore, option (c) is the correct answer.
Relevance: The Wildlife Crime Control Bureau (WCCB) of the Ministry of Environment, Forest and Climate change has constituted a team to investigate the death of ten elephants in Bandhavgarh Tiger Reserve of Madhya Pradesh.
Subject: Environment | Biodiversity Conservation
Tags: Factual

Q50.
Answer: d
Explanation:

- **Kwango province** is in the **Democratic Republic of the Congo** (DRC). It was recently in the news due to a mysterious illness called "**Disease X**". Later it was found to be a **severe form of malaria**.
- The term **"Disease X"** is frequently used by the World Health Organization (WHO) and other public health bodies as a placeholder for a new, potentially severe pathogen that could cause significant morbidity and mortality before its etiology is clearly understood.
- Between 24 October and 5 December 2024, **Panzi health zone in Kwango Province** of DRC recorded **406 cases of an undiagnosed disease** with symptoms of fever, headache, cough, runny nose and body ache. All severe cases were reported to be severely malnourished. Among the cases, 31 deaths have been registered. The majority of cases reported are among children, particularly those under five years of age.

Therefore, option (d) is the correct answer.
Subject: Polity | International Relations
Tags: Factual

Q51.
Answer: a
Explanation:

- **Tropical cyclones** are warm-core low pressure systems, which do not have any attached front, developing over the tropical or subtropical waters, they have an organized circulation.
- These are storm systems which gain energy from latent heat, that is from water vapor condensation. The development and the intensity of these cyclones depend on high underlying sea surface temperatures (at least 26.5 °C in the upper most ocean layer), a strong decrease of air temperature with altitude and low wind shear in the upper troposphere These factors permit strong convection and condensation. The initialization of a cyclone requires a minimum of Coriolis force and a pre-existing disturbance in atmospheric circulation.
- The **North Indian Ocean cyclone season has two peaks** – in the **pre monsoon months** of April and May, and the **post monsoon months** of mid-October to December, with the monsoon months reporting no or less cyclonic activity because of high vertical wind shear.
- Even amongst the pre-monsoon and post-monsoon months, it is the latter which have the most cyclonic activity. **So, statement 1 is correct**.
    - This is because summers and **pre-monsoons see dry and hot north-westerly winds over north-west India which moves towards the Bay of Bengal.** This blocks the rise of air from the water, and the subsequent formation of clouds, preventing cyclone-friendly conditions. But the absence of this air movement in the post-monsoon phase increases the chances of cyclones. **So, statement 2 is correct.**

So, both statement I and statement II are correct, with statement II correctly explaining statement I.
Therefore, option (a) is the correct answer.
Relevance: Cyclone Fengal made landfall on the coast of Puducherry
Subject: Geography | Climate Weather and Atmosphere
Tags: Conceptual, Analytical

Q52.
Answer: b
Explanation:

- **Information Technology (IT) Act 2000,** empowers the governments with the authority to issues directives for
  - **Interception** : Interception electronic communication passing through or being processed in a computer system. **So, statement I is not correct.**
  - **Monitoring** : To monitor or view electronic communications passing through or being processed in a computer system
  - **Decryption** : to decrypt encrypted electronic communications stored or transmitted through a computer system.
- Under the **Information Technology(IT) Act 2000**, the Ministry of Electronics and Information Technology (**MeitY**) has the authority to **block access to websites and online content.** This authority is exercised under **Section 69A** of the Act which allows the government to issue orders to intermediaries such as Internet Service Providers (ISPs), telecom operators, web hosting platforms, and search engines. **These orders are issued when the content is deemed as a threat to India's sovereignty, integrity, national security, or public order. So, statement 2 is correct.**
- **Section 69** of the IT Act allows the government to issue content-blocking orders to online intermediaries such as Internet Service Providers (ISPs), telecom service providers, web hosting services, search engines, online marketplaces, etc.
  - As per the act, the intermediaries are the entities that facilitate the transmission, storage or retrieval of electronic records, and includes ISPs, and engines, social media, websites and ecommerce portals.

Therefore, option (b) is the correct answer.
Subject: Polity | Governance
Tags: Factual

Q53.
Answer: b
Explanation:

- Ferecrystals are a unique class of **Misfit Layered Compounds (MLCs)** consisting of at least two or more periodically stacked independent layers.
- These layers exhibit **weaker bonding along the stacking direction** due to a misalignment between the repeating patterns of the two layers, causing a 'misfit.'
- The **twisting between the layers (rotational disorder)** in ferecrystals hampers heat transport significantly, effectively blocking heat waves.
- This exceptional property makes ferecrystals particularly suitable for **thermoelectric energy conversion**, enabling the conversion of waste heat into electricity.

- Ferecrystals possess a **high thermoelectric figure of merit**, exceeding the value 2, indicating their outstanding performance in thermoelectric applications.
- Incorporating ferecrystals as **nanostructures in a solid-state matrix** is a challenging task in synthetic chemistry and materials science but holds promise for significant advancements in thermoelectric technology.
- Other Options:
    - **Quasicrystals**: These are crystal-like substances, but unlike a crystal in which atoms are arranged in a repeating pattern, atoms in quasicrystals consist of atoms that do not repeat themselves regularly.
    - **Quantum Dots**: These are nanometer size crystals with semiconducting properties. The underlying principle behind quantum dots is that properties of any element depend on how many electrons it has. However, when matter is really tiny, the properties depend on its size.
    - **Vitrimers**: Substances with the capacity of being reprocessed and reshaped multiple times without breaking its structure.

Therefore, option (b) is the correct answer.
Relevance: Researchers have recently developed a new material with potential for highly efficient conversion of waste heat to energy by introducing twisted layers in ferecrystals.
Subject: Science & Technology | Miscellaneous
Tags: Conceptual

Q54.
Answer: a
Explanation:

- Atmospheric river (AR) is a term used for **narrow, flowing bands of condensed water vapor in the atmosphere** that transport moisture from the tropics to higher latitudes. While many atmospheric rivers are weak systems that bring beneficial rain or snow, larger and more powerful ones can cause extreme rainfall and floods, resulting in catastrophic damage.
- These are **pole-bound streams** of water vapor fed by the warming oceans in the tropics. **So, statement 1 is not correct.**
- ARs can sometimes carry many times the volume of moisture and flow at a higher speed than some of the world's major rivers. They **hover much closer to land than the dry jet streams** and pick up moisture moving over the vast, warm oceans, which they shed while traversing over land. **So, statement 2 is correct.**
- As they travel north, they are propelled by strong winds and **bring heavy rainfall** to areas in their path. **So, statement 3 is not correct.**
- In a warming world, these **atmospheric rivers are becoming longer, wider and more intense,** primarily due to greater availability of water vapor and can cause massive destruction and loss of life. **So, statement 4 is not correct.**

So, only one of the given statements is correct.
Therefore, option (a) is the correct answer.
Subject: Environment | Climate change
Tags: Conceptual

Q55.
Answer: a
Explanation:

- **Hydrothermal vents** are underwater hot seeps that form in areas of active volcanoes and seamounts. Most are **found along continental plate boundaries.** Hydrothermal vents are home to primary producers that obtain their energy from volcanic gases, and have also been found to act as a recycling and decomposition system for dissolved organic carbon, which is essential in the context of the global carbon pool. These vents support diverse ecosystems and microbial communities, serving as the foundation of the food web in deep-sea environments.
- Hydrothermal vents are found in the **global mid-oceanic ridge system**, which form when plates of the **Earth's lithosphere move apart**, giving rise to a new seafloor. As the plates separate, magma rises above the seafloor in the form of a continuous range of underwater volcanic eruptions. **So, statement 1 is not correct.**
- Hydrothermal vents support a high gradient of biodiversity, living within and around the vents hundreds of species including crabs, shrimps, fish, and octopus as well as sessile creatures such as barnacles, limpets, feather stars, and tube worms. **Due to the depth and nature of hydrothermal vents, many of these species are endemic** (only within that area). These support greater biodiversity due to the **habitat enrichment and bacterial symbiosis from the vent fluids**. They are largely considered as the sites for the origin of life. **So, statement 2 is not correct.**
- Within the hydrothermal vents are **seafloor massive sulfides (SMS)**, whereby the vents create sulfide deposits containing valuable metals such as silver, gold, manganese, cobalt, and zinc. **So, statement 3 is correct.**

So, only one of the above given statements is correct.
Therefore, option (a) is the correct answer.
Relevance: Indian oceanographers have captured the image of an active hydrothermal vent located 4,500 metres below the surface of the Indian Ocean.
Subject: Environment | Biodiversity Conservation
Tags: Factual

Q56.
Answer: b
Explanation:

- The **Chishti order** of Sufism is one of the most prominent Sufi orders in the Indian subcontinent. It emphasized simplicity, renunciation of power, and spiritual devotion. The Chishti saints were known for their independence from rulers, their cultural assimilation, and their alignment with local traditions, which sometimes led to conflict with orthodox Islamic views.
- The Chishti saints **deliberately avoided close relations with rulers and political authorities,** believing that **wealth and power distracted from the pursuit of spirituality.** For instance, **Nizamuddin Auliya** refused gifts and grants from the Sultanate and even redistributed the money given by the Sultan to the poor. Their focus was on serving the marginalized, not on aligning with state power. **So, statement 1 is not correct.**
- The Chishti order was a ***Ba-shara*** tradition, meaning it adhered to Islamic law (Sharia). While they incorporated local practices and emphasized tolerance, they never rejected Islamic principles. This distinguishes them from *Be-shara* orders, which often rejected legalistic religious frameworks.

- The Chishti order embraced **local traditions** like devotional music (**sama**) and practices that resonated with the Indian populace. Sufi traditions especially those of the Chishti order of Sufism to which **Moinuddin and later Sufi masters such as Qutbuddin Bakhtiyar Kaki, Hazrat Nizamuddin Auliya, and Nasiruddin Mahmud Chiragh Dehlavi,** all of Delhi, belonged drew heavily from pre-existing local practices, which were seen as heretical in orthodox Islam. However, this approach helped the Chishti order gain immense popularity in India.
- In the same line Chishti saints, such as **Nizamuddin Auliya**, maintained contact with the **Nath Panth**, a sect with significant influence during that time. They learned **yogic practices**, reflecting their ability to assimilate local traditions into their teachings, thus building a connection with Indian culture. This is the reason why Nizamuddin Auliya was referred to as a *siddh yogi*. **So, statement 2 is correct.**

Therefore, option (b) is the correct answer.
Subject: Art and Culture | Indo Islamic Culture
Tags: Conceptual

Q57.
Answer: b
Explanation:

- Sickle Cell Disease (SCD) is an inherited disease characterized by abnormal red blood cells that take on a crescent or sickle shape. Irregularly shaped cells can obstruct blood vessels, resulting in a range of health complications.
- **Sickle cell disease (SCD) is an inherited blood disorder.** That means it is passed down from a parent's genes. It causes the body to make abnormal hemoglobin. **So, statement 1 is not correct.**
    - It is megaloblastic anemia which occurs because of Vitamin B12 deficiency.
- The **Ministry of Health and Family Welfare (MoHFW) has identified SCD** as one of the ten major health issues that **disproportionately affect the tribal population in India.**
- The MoHFW has launched a **National Sickle Cell Disease Anaemia Elimination Mission in 2023 to prevent and manage hemoglobinopathies and to eliminate the sickle cell burden in India by 2047.** The mission aims to promote health, screen individuals below 18 years, and focus on early detection of cases through population-based screening and providing holistic care to individuals with Sickle cell disease. **So, statement 2 is correct.**

Therefore, option (b) is the correct answer.
Subject: Science & Technology | Health and Diseases
Tags: Conceptual

Q58.
Answer: b
Explanation:

- **Direct money transfer schemes** such as **Ladli Behena and Lakshmir Bhandar**, are designed to **provide financial support to** individuals and families. These schemes **aim to reduce poverty, promote financial inclusion, and improve the overall well-being** of marginalized populations. However, along with the intended benefits such schemes can also have some **undesirable consequences** that may negatively impact the economy.
- It may lead to **unmindful consumption**, which might not always be productive or efficient and could lead to **overexploitation of resources**. The **Economic Survey of India 2021-22** had mentioned

that when households receive money without a direct link to investment or long-term productivity, there is **a risk of spending on non-essential items or low-value goods like alcohol**. This results in **increased wasteful expenditure. So, point 1 is correct.**

- These schemes significantly **increase the government expenditure** as it requires substantial funding for financing the direct money transfer. The increase in the government expenditure can be financed either through taxes or other sources. **If government revenues through taxes or other sources do not rise in proportion**, the fiscal deficit expands. This increase in the **deficit may lead to increased borrowing,** which can crowd out private investment and put **pressure on inflation. So, point 2 is correct.**
- Bond Yields are directly proportional to inflation rates in the economy, with increased demands for funds by the government the **bond yields will further increase. So, point 3 is not correct.**
- Direct money transfers **enhance the purchasing power of individuals** by providing them with additional income. This additional income **enables better access to goods and services** which can improve the quality of life for many beneficiaries. **So, point 4 is not correct.**

So, only two of the above given points are correct.
Therefore, option (b) is the correct answer.
Subject: Economy | Public Finance
Tags: Conceptual

Q59.
Answer: d
Explanation:

- **The Bharatiya Vayuyan Vidheyak, 2024** replaced the Aircraft Act, 1934, and continues the regulatory framework for India's civil aviation sector.
- **The Act grants statutory recognition to three key authorities:**
  - **Directorate General of Civil Aviation (DGCA)**: Responsible for regulatory functions and overseeing aviation safety. **So, statement 1 is correct.**
  - **Bureau of Civil Aviation Security (BCAS):** Oversees security aspects within civil aviation.
  - **Aircraft Accidents Investigation Bureau (AAIB):** Investigates aircraft accidents.
- The Act allows the Central Government to impose penalties for contravention of rules. **It empowers the Central Government to appoint an officer for adjudication of penalties.** Decisions of the Adjudicating Officer may be appealed before an Appellate Officer. The appeals from appellate officers can further be made with a second appellate officer. **So, statement 2 is correct.**
- The Act allows the **Central Government** to impose penalties for contravention of rules. It empowers the Central Government to **appoint an officer for adjudication of penalties.** This officer must be of the rank of the **Deputy Secretary** to Government of India or above. Decisions of the Adjudicating Officer may be appealed before an Appellate Officer, who must be of a higher rank than the Adjudicating Officer.
  - The Act specifies that **appeals against orders issued by the DGCA or BCAS** are to be made to the Central Government**.** Importantly, it states that **no further appeals** are permitted beyond this level, making the Central Government's decision final in such cases. **So, statement 3 is correct.**

Therefore, option (d) is the correct answer.

Relevance: The Bharatiya Vayuyan Vidheyak, 2024 was passed by the parliament in the recently concluded winter session of parliament.
Subject: Economics | Economic Schemes and Policies
Tag: Factual

Q60.
Answer: **a**
Explanation:

- Acquired Immunodeficiency Syndrome (AIDS) is a life-threatening condition which is caused by the Human Immunodeficiency Virus (HIV). HIV is a **retrovirus** that primarily targets the immune system leading to a weakened immune response. As the virus replicates it severely compromises the body's ability to fight infections and diseases.
- HIV is classified as a **retrovirus** as it carries its genetic material in the form of **single-stranded RNA (ssRNA)**. Unlike double-stranded DNA viruses, HIV must first convert its RNA genome into DNA to replicate in host cells. This process is facilitated by an enzyme called **reverse transcriptase** which is unique to retroviruses.
- **The reverse transcriptase enzyme is highly inefficient and often makes mistakes when copying RNA into DNA. These mistakes create new versions of the virus**. Leading to the virus changing rapidly because of its high mutation rate. As a result of which HIV has many types and subtypes. For example, HIV-1 has subtypes like A, B, and C. **So, statement 2 is correct.**
- Vaccines work on the principle of training the immune system for an imminent attack from the causative organism, for this purpose a non-virulent form of virus, structurally similar to the virulent is introduced in the body for making the immune system recognize the same. As the reverse transcriptase is an inefficient enzyme it leads to large scale variabilities in successive generations of viruses, thus **making it difficult to create a vaccine for AIDS. So, statement 1 is correct.**

So, both Statement I and Statement II are correct, Statement II explains Statement I.
Therefore, option (a) is the correct answer.
Subject: Science & Technology | Health and Diseases
Tags: Conceptual, Analytical

Q61.
Answer: c
Explanation:

- Tiger reserves are notified under the Wildlife Protection Act, 1972. These consist of extensive areas of land that include both a core zone and a buffer zone. The core area is designated as a national park or sanctuary, providing strict protections for wildlife. In contrast, the buffer zone is a mix of forested and non-forested land that serves various uses. These buffer zones act as transitional areas that support wildlife movement and habitat.
- The state of **Madhya Pradesh** has the **highest number of Tiger reserves** in India **(eight tiger reserves)**. They are Kanha, Satpura, Bandhavgarh, Pench, Sanjay Dubri, Panna, Veerangana Durgavati and Ratapani Wildlife Sanctuary. **So, statement 1 is correct.**
- Tiger Reserves are **notified by State Governments** as per provisions of Section 38V of the **Wildlife (Protection) Act, 1972** on **advice of the National Tiger Conservation Authority. So, statement 2 is not correct.**

- As per section 38W of the Act, once the notification comes "n**o State Government shall de-notify or alter the boundaries of a tiger reserve,** except in public interest with the approval of the Tiger Conservation Authority and the National Board for Wild Life". **So, statement 3 is correct.**

Therefore, option (c) is the correct answer.
Relevance: Ratapani Wildlife Sanctuary was recently declared the 8th Tiger Reserve of Madhya Pradesh.
Subject: Environment | Biodiversity Conservation
Tags: Factual

Q62.
Answer: c
Explanation:

- The **Tishrin dam is located in Syria** on the west bank of the Euphrates River. approximately 70% of Syria's electricity needs are met from the Tishrin, Tabqa and Baath dams. The Tishrin dam has been a target in the Syrian Civil war.
- Also, it is strategically important as the Euphrates river is the only crossing between Turkey and the Kurdish part of Syria.

Therefore, option (c) is the correct answer.

Knowledge Box

Important areas in Middle East:
• **Orontes River**
  o **Location:** The river flows through Syria, Lebanon, and Turkey and empties into the Mediterranean Sea.
  o **Significance:** It is vital for farming, industry, and daily use. It has historical and cultural importance and also linked to ancient civilizations. But, over 12 years of war in Syria have damaged these resources and degraded the water in river.
• **Golan Heights**
  o **Location:** A plateau region in southwestern Syria which is bordered with Lebanon, Israel, and Jordan.
  o **Significance:** A strategically important area due to its elevation, control over water resources and agricultural capacity. It has been a subject of conflict between Israel and Syria for decades. Israel captured the Golan Heights in the 1967 Six-Day War and later annexed it.
• **Barada River**
  o **Location:** The river flows through the Damascus (Capital of Syria)
  o **Significance:** It is a vital water source for Damascus and its surrounding areas. However, water scarcity due to climate change, population growth, and mismanagement has posed challenges to the river's flow and the region's water security. For the last few years the Damascus administration has been trying to recover the dying river.

Subject: Polity | International Relations
Tags: Conceptual

Q63.
Answer: b
Explanation:

- The **New Development Bank (NDB)** is a multilateral development bank **established by the founding members of the BRICS grouping (Brazil, Russia, India, China, and South Africa)** with the primary goal of mobilizing **funds for infrastructure and sustainable development** projects in emerging markets and **developing economies. So, statement 1 is correct.**
- **Membership** in NDB is **open to any member of the United Nations.** Current members include the following:
    - **Founding members:** Brazil, Russia, India, China, and South Africa.
    - **New members:** Bangladesh, the United Arab Emirates, Uruguay, Egypt, and Algeria
- While the **five founding BRICS members** had **equal shareholding of 18.45% each**, **new members** like Bangladesh, UAE, Egypt, and Algeria have **varying shares.** For instance, **Bangladesh has 1.74%,** Algeria has 2.77%, and **UAE has 1.03%.** The addition of new members has altered the equal shareholding concept, though the BRICS group must maintain at **least 55% of total shares. So, statement 2 is not correct.**
- **Governance Structure of NDB:**
    - The Bank is governed by a **Board of Governors** made up of the finance ministers of the five BRICS countries, and a Board of Directors.
    - **Voting power** within the Board is based on each country's **shares in the bank.**
    - **The NDB president** is elected on a **rotational** basis **from one of the founding members,** and there are four vice presidents from each of the other four founding members. **So, statement 3 is correct.**

Therefore, option (b) is the correct answer.
Relevance: India has contributed $2 billion to the BRICS New Development Bank (NDB) in seven installments between 2015-16 and 2021-22.
Subject: Polity | International Relations
Tags: Factual

Q64.
Answer: d
Explanation:

- A **money mule** is someone who transfers or moves illegally acquired money on behalf of another party. Criminals recruit money mules to help launder proceeds derived from online scams, frauds, or other crimes.
    - **A mule account is a bank account** used by criminals to launder illicit funds, often set up by **unsuspecting individuals lured by promises of easy money** or coerced into participation. The transfer of funds through these highly interconnected accounts makes it difficult to trace and recover the funds.

Therefore, option (d) is the correct answer.
Relevance: The Reserve Bank of India (RBI) has introduced MuleHunter.ai, an AI/ML-based tool designed to detect and neutralize mule bank accounts.
Subject: Economy | Money and Banking
Tag: Analytical

Q65.
Answer: b
Explanation:

- **The Rights of Persons with Disabilities Act, 2016** provides for the **right to free education** for children with **benchmark disabilities** (those with at least 40% disability) between **6 and 18 years.** It mandates government-funded and recognized educational institutions to provide **inclusive education** to ensure equal opportunities for children with disabilities. **So, statement 1 is correct.**
- Under the Rights of Persons with Disabilities Act, 2016, only the **Central Government has the power to add any new category of specified disability.** This ensures uniformity across the country and aligns with India's obligations under the United Nations Convention on the Rights of Persons with Disabilities (UNCRPD). **So, statement 2 is not correct.**
- **The Accessible India Campaign (Sugamya Bharat Abhiyan):** Launched in 2015, it aims to achieve **universal accessibility for Persons with Disabilities (PwDs)** across India. The key focus areas include improving Built Environment Accessibility in public spaces, enhancing Transportation Accessibility for independent mobility, creating an accessible **Information and Communication ecosystem,** and expanding **Sign Language Access** through interpreter training and better media support. **So, statement 3 is not correct.**
- **Deendayal Divyangjan Rehabilitation Scheme (DDRS):** It is a central sector scheme to provide **grant-in-aid to non-governmental organizations (NGOs)** for projects relating to the **rehabilitation of persons with disabilities** aimed at enabling persons with disabilities to reach and maintain their optimal, physical, sensory, intellectual, psychiatric, or socio-functional levels. **So, statement 4 is correct.**

So, only two of the above given statements are correct.
Therefore, option (b) is the correct answer.
Relevance: The International Day of Persons with Disabilities (IDPD), observed annually on December 3, celebrates the resilience, contributions, and leadership of persons with disabilities (PwDs) worldwide.
Subject: Polity | Governance
Tags: Factual

Q66.
Answer: b
Explanation:

- The **cheetah (*Acinonyx jubatus*)** is a **large cat** native to **Africa** and **central Iran**. It is the fastest land animal, capable of reaching speeds up to 120 km/h in short bursts. Cheetahs are **carnivores** primarily preying on small antelopes.
- The **cheetahs** are classified as **"Vulnerable"** on the **IUCN Red List. So, statement 1 is not correct.**
- **Cheetahs** are primarily adapted to **open grasslands, savannahs, and semi-arid** landscapes. In Africa, they inhabit vast open plains. India's grasslands, like those in **Kuno National Park,** have been selected to match the Cheetah's natural habitat. Cheetahs **cannot thrive in dense forests,** which are unsuitable for their speed-based hunting techniques. **So, statement 2 is correct.**

Therefore, option (b) is the correct answer.
Relevance: Project Cheetah completed two years in 2024 since the first batch of cheetahs arrived in India from Africa in September 2022.

Subject: Environment | Biodiversity and Conservation
Tags: Factual

Q67.
Answer: a
Explanation:

- **India's Total Fertility Rate**, as per National Family Health Survey-5 (2019-21), has dropped to **2.0**, which is below the replacement level of **2.1**. The replacement level indicates the rate at which a population exactly replaces itself from one generation to the next, without migration. A TFR below 2.1 signals population stabilization, a key milestone in India's demographic transition. **So, statement 1 is correct.**
- As per the data of the Sample Registration System (SRS) Statistical Report 2020, general fertility rate (GFR) reveals that **fertility in all the age groups is higher in rural areas than in urban areas**.
- Estimates of General Fertility Rate (GFR), a refined measure of fertility, has been defined as the number of live births per thousand women in the reproductive age group 15-49 years
    - Fertility rates have traditionally been higher in rural areas due to limited access to healthcare, education, and family planning services.
    - Although fertility rates are declining in rural areas, the pace is generally **slower** compared to urban areas, where better services and awareness have led to quicker reductions in fertility. **So, statement 2 is not correct.**
- India was the first country in the world to adopt an official population policy and launch an official family planning program way back in 1952. Since them, planned efforts have been the mainstay of population stabilization program, with population control being a mainstay on the erstwhile five year plans since the first five year plan of 1951-56, but the targets were largely based on the societal and moral change, than the adoption of an objective criterion.
- It was only by the **Fourth Five Year Plan (1969-74)** that for the first time, set up targets in terms of sterilization and Intrauterine Contraceptives insertions and to widen the acceptance of oral and injectable contraceptives in order to achieve the aim of reducing the birth rate to about 32 per thousand population by 1973-74 from the birth rate of 39 per thousand population**. So, statement 3 is not correct.**

So, only one of the above given statements is correct.
Therefore, option (a) is the correct answer.

Knowledge Box
India's five year plans and Population control: • **First Five-year Development Plan (1952-57)**: The plan recognized that the increase in population and the pressure exercised on India's limited resources had brought to the forefront the urgency of the problem of family planning and population control. The Plan argued that application of medical knowledge and social care had lowered the death-rate, while the birth-rate remained fairly constant which had led to rapid increase in the growth of population. **The Plan advocated that this could be secured only by the realization of the need for family limitation on a wide scale by the people.** The Plan however insisted that the main appeal for family planning was based on considerations of the health and welfare of the family. • **Second Five-year Development Plan (1957-61)**: This plan proposed a program for family limitation and population control. The Plan suggested that this program

should, among others, make advice on family planning an integral part of government hospitals and public health agencies.

- **Third Five-year Development Plan (1961-66)**: It emphasized that the greatest stress should be placed on the program of family planning and that the objective of stabilizing the growth of population over a reasonable period must be at the very center of planned development. The emphasis during the Plan was on expanding the availability of family planning services, especially sterilization services within the public health care delivery system.
- **Fourth Five-year Development Plan (1969-74):** It signaled the domination of the demographic rationale of family planning over its health rationale. **The Plan, for the first time, set up targets in terms of sterilisation and IUCD insertions and to widen the acceptance of oral and injectable contraceptives in order to achieve the aim of reducing the birth rate to about 32 per thousand populations by 1973-74 from the birth rate of 39 per thousand populations that prevailed at that time.** This was the beginning of the target approach that dominated the planning and implementation of planned family planning efforts in India for the next 35 years.
- **Fifth Five-year Plan (1974-79)**: It also gave topmost priority to family planning. The period 1974-79 was also a period of political turbulence in India. Emergency was clamped in 1975 and a major push was given to family planning, especially sterilisation, during 1975-77. This push brought in elements of coercion and force in planned family planning efforts.
- **The Sixth Five-year Development Plan (1980-85):** It attempted to bring the planned family efforts at the centre stage of the development agenda of the country and aimed at the long-term demographic goal of reducing the net reproduction rate (NRR) to one by 1996 for the country as a whole and by 2001 in all the States from the NRR of 1.67 that prevailed at that time.
- **The Seventh Five-year Plan (1985-90):** This plan targeted a couple protection rate of 42 per cent to bring down the birth rate to 29.1 per thousand population along with targets in terms of reduction in the death rate, infant mortality rate and improvements in the coverage of child immunisation and ante-natal care.
- **Eighth Five-year Plan (1992-97)**: It aimed at reducing the birth rate from 29.9 per thousand in 1990 to 26 per thousand by 1997. The Plan stressed the need of a National Population Policy and suggested an inter-sectoral approach supported by political commitment and popular mass movement.
- **Ninth Five-year Development Plan (1997-2002)**: The rationale for planned family planning efforts was confined to meeting the felt-needs of family planning of eligible couples. Planned family planning efforts were completely subsumed in the reproductive and child health efforts resulting in a substantial dilution of family planning efforts.
- **Tenth Five-year Plan(2002-07)** called for the integration of numerous vertical programmes for family planning and maternal and child health into integrated programme of health care for women and children; a shift from demographic targets to

enabling couples to achieve their reproductive goals; and meeting all unmet needs of contraception to reduce unwanted pregnancies.

Relevance: A political leader recently made comments on the concerns in regard of the falling fertility rates in Indian society.
Subject: Economics | Population and Demography
Tags: Factual

Q68.
Answer: c
Explanation:

- The term **'Silicon Shield'** refers to **Taiwan's strategic deterrence** against potential aggression due to its **global dominance in semiconductor manufacturing.** Taiwan, led by Taiwan Semiconductor Manufacturing Company, produces over **90% of the world's most advanced chips**, which are critical for devices like smartphones, AI servers, and military systems.
- Any disruption to Taiwan's semiconductor production would cause **catastrophic consequences for the global economy**, forcing major powers like the U.S. to intervene to secure supply chains.
- While this manufacturing prowess deters aggression, it **also increases risks**, as Taiwan becomes a focal point of **geopolitical competition,** particularly between China and the U.S.

Therefore, option (c) is the correct answer.
Relevance: Taiwan's critical role in global semiconductor production has gained focus amid rising U.S.-China tensions and efforts to secure supply chains.
Subject: Polity | International Relations
Tags: Factual

Q69.
Answer: a
Explanation:

- **The Martand Sun Temple,** located in Anantnag and built during the **Karkota Dynasty** under Lalitaditya Muktapida (8th century AD), is noted for a **unique blend of Gandhara, Chinese, and Gupta styles**. The temple has a **pyramidal roof** and **elaborate carvings of deities**, including the Sun God, Vishnu, and river goddesses. **So, statement 1 is correct.**
- **Islamic architecture** in Kashmir under the S**hahmiri and Mughal rulers** adopted local elements. Mosques **lacked the traditional domes and cloisters** found elsewhere in India. Instead, **multi-tiered sloping roofs, wooden pillars**, and **tall spires were preferred,** as seen in monuments like Jamia Masjid and Aali Masjid. **So, statement 2 is not correct.**
- Mughal rulers, inspired by P**ersian designs,** constructed **gardens in the Char Bagh style.** Features like terraces, water streams, fountains, and lush greenery were integral. Prominent gardens include **Shalimar Bagh (built by Jahangir), Nishat Bagh**, and **Chashme Shahi. So, statement 3 is correct.**

Therefore, option (a) is the correct answer.
Relevance: J&K Chief Secretary recently approved the inclusion of religious places, including mosques, under the PM Surya Ghar Muft Bijli Yojana to enhance electricity generation in the power-deficient region.
Subject: History | Architecture and Sculptures
Tags: Factual

Q70.
Answer: **a**
Explanation:

- The Inter-Tropical Convergence Zone, or ITCZ, is the region that circles the Earth, near the equator, where the trade winds of the Northern and Southern Hemispheres come together. The intense sun and warm water of the equator heats the air in the ITCZ, raising its humidity and making it buoyant. Aided by the convergence of the trade winds, the buoyant air rises. As the air rises it expands and cools, releasing the accumulated moisture in an almost perpetual series of thunderstorms.
- **ITCZ** is a band of intense rainfall that encircles the earth and is responsible for 32 per cent of the world's precipitation. **It moves north and south across the equator over the course of a year following the seasonal cycle of temperature differences between the hemisphere.**
- Recent studies have indicated that **ITCZ is getting narrower and shifting northwards**, with the recent event of severe rainfall in the Sahara being a consequence of this change only. **So, statement 1 is correct.**
- One of the major reasons for ITCZ shifting northwards than it shifts to south is that the Northern hemisphere has a larger land area than the southern hemisphere, as the area gets heated at much larger extent there is **increased convection available to feed in the ITCZ** thus pushing it further north. **So, statement 2 is correct.**

So, both Statement–I and Statement–II are correct and Statement–II explains Statement–I.
Therefore, option (a) is the correct answer.
Subject: Environment | Climate change
Tags: Conceptual

Q71.
Answer: c
Explanation:

- **Renewable Purchase Obligations (RPOs**) are **statutory mandates under the Electricity Act, 2003,** requiring obligated entities like distribution companies, open access consumers, and captive power producers to source a specified percentage of their electricity from renewable sources. The **National Tariff Policy, 2016,** further supports this framework. **So, statement 1 is not correct.**
- The **Ministry of Power**, **not the Ministry of New and Renewable Energy** (MNRE), sets the RPO trajectory at the **national level**. For instance, the Ministry of Power has issued orders detailing the RPO and Energy Storage Obligation trajectory till 2029-30. **So, statement 2 is not correct.**
- **State Electricity Regulatory Commissions (SERCs)** are responsible for enforcing RPO compliance within their jurisdictions. They set state-specific RPO targets and monitor adherence by obligated entities. **So, statement 3 is correct.**
- **Compliance Mechanism:** Obligated entities can meet RPOs by purchasing renewable energy directly or through Renewable Energy Certificates (RECs).
- **Penalties for Non-Compliance:** SERCs may impose penalties on entities failing to meet RPO targets, which can include fines or directives to purchase additional RECs.

Therefore, option (c) is the correct answer.
Relevance: Only five states have consistently met their solar RPOs — Karnataka, Andhra Pradesh, Gujarat, Punjab and Uttar Pradesh.

Subject: Economics | Sustainable Development and Growth
Tags: Factual
Q72.
Answer: d
Explanation:

- **Atmospheric blocking** is the formation of a quasi stationary vortex or vortex pair over mountains which is sufficiently strong to block the usual mid-latitude storm track, forcing weather systems to detour around the block.
- **It occurs when a jet stream ridge becomes especially large and forms a separate anticyclonic eddy in the flow**, which creates a large-scale persistent weather pattern that "blocks" the propagation of Rossby waves.
- A recent study by Indian Institute of Scientific Education and Research, Mohali has noted that an **atmospheric block over Central Asia has effectively blocked the movement of warmer westerlies,** preventing them from moderating the cold temperatures in North India. Thus, leading to the prolonged Cold Wave conditions in India.

Therefore, option (d) is the correct answer.

Knowledge Box:

- **Cold wave** is defined as an extreme weather event marked by a negative departure **(a drop in average temperature)** well below the average for that region for a sustained period of two days or more.
- India's core cold wave zone covers the northern, central and eastern parts. But the west and south come in for spells too.
- **According to the Indian Meteorological Department, a cold wave occurs when the minimum temperature is 10°C or less for plains and 0°C or less for the hilly regions** and negative departure from normal day temperature is 4.5 to 6.4°C. If negative departure is more than 6.4°C, it is a severe cold wave. Which means that day temperatures play an important role in cold waves.
- Additionally, the plains suffer a cold wave when minimum temperature is 4°C or less and a severe cold wave when minimum temperature falls to 2°C or less. Coastal areas suffer cold waves when the minimum temperature is 15°C or less and the negative departure from the minimum is 4.5 °C.
- **Cold waves in India were considered to be influenced by 'Western disturbances' or cold winds starting from the Mediterranean across the Caspian Sea and blowing into India across Afghanistan-Pakistan.**
- A phenomenon called '**Arctic amplification**' — coming all the way from the Arctic and Siberia — descends upon India to cause extreme cold events.
- **The global weather phenomenon 'la Nina', with its cold winds all the way above the Eastern Pacific, also influences cold wave conditions across India.**

Relevance: The National Capital Territory Delhi recently experienced five cold waves, with its nights being the coldest since 2013 and days since 2003.
Subject: Geography | Climatology

Tags: Conceptual

Q73.
Answer: b
Explanation:

- **Copper,** known for its high conductivity, malleability, and corrosion resistance, plays a crucial role in modern industries.
    - Itis **malleable and ductile**, Resistant to corrosion
    - It is **100% recyclable** without any loss in quality, making it an eco-friendly metal.
- **India is not self-sufficient** in refined copper production. India **imports 30-40% of its refined copper,** with Japanese suppliers accounting for 80% of imports. **So, statement 1 is not correct**.
- Globally, most commercial copper ore deposits have an average grade of 0.8% copper. In contrast, India's copper ore has an average copper content of around 1%, which is slightly higher than the global average. **So, statement 2 is correct.**
- **Rajasthan is the leading State in copper production in India**, followed by Madhya Pradesh and Jharkhand. The **Khetri copper complex** in Rajasthan is one of the major producers. **So, statement 3 is correct.**

Therefore, option (b) is the correct answer.
Relevance: The Quality Control Order (QCO) on refined copper, effective from December 1, has barred imports from Japanese suppliers, who account for 80% of India's copper imports, as they await BIS certification.
Subject: Geography | Economic geography
Tag: Conceptual, Factual

Q74
Answer: d
Explanation:

- **Attribution studies** calculate whether, and by how much, climate change affected the intensity, frequency or impact of extreme weather events.
- In these types of study, scientists precisely measure the impact of climate change on a weather event, or assess if the particular event would have been different, both in frequency and intensity without climate change.

Therefore, option (d) is the correct answer.
Subject: Environment | Climate change
Tags: Conceptual

Q75.
Answer: b
Explanation:

- **Potatoes** can be grown only under such conditions where the **temperatures during the growing season are moderately cool.**
    - In India the planting time for the crop varies from region to region, and is cultivated as both Kharif and Rabi crops in different regions. **So, statement 1 is not correct.**
    - The optimum temperature for vegetative growth is 24°C, while the ideal temperature for tuber development is 20°C. Temperatures above 30°C can stop bulb formation.

- The potato can be grown almost on any type of soil except saline and alkaline soils.

- Potatoes are grown in **almost all States** of India. **Uttar Pradesh** followed by West Bengal were the major producers in 2021-22. These **two States** accounted for **almost 50%** of India's total production. **So, statement 2 is correct.**
- **Operation Greens** is a Central Sector Scheme which aims to promote Farmer Producers Organizations (FPOs), agri-logistics, processing facilities and professional management for certain agricultural commodities **including Potato**. These interventions from the Government are mainly focused on maintaining a steady supply of the commodity and preventing price fluctuations in the market. **So, statement 3 is correct.**
- Potato prices in India are subject to seasonal fluctuations due to varying supply levels, with prices typically rising during lean periods and falling during harvest seasons.

Therefore, option (b) is the correct answer.

Relevance: Since the onset of December 2024, the West Bengal government has stopped the export of potatoes to its neighboring states, causing the price of the vegetable to shoot up there

Subject: Economics | Agriculture and Food Management

Tag: Factual

Q76.

Answer: d

Explanation:

- **Dharini** is a **brain atlas** developed by the researchers at **Indian Institute of Technology, Madras.** This brain atlas is first of a kind and the only in the world to capture the developing human brain at the fetal stage. It uses advanced technology to map over 5,000 brain sections and more than 500 brain regions. As announced the data from this atlas would be available in public domain, thus facilitating further research.
- The **only other publicly available brain atlas such as this was released by US Allen Institute for Brain Science** in 2016. It captured the brain of an adult woman in 1,356 plates.

Therefore, option (d) is the correct answer.

Knowledge Box
• **BHAVINI:** It is a new public sector undertaking of the Government of India to manage Fast Breeder Reactors. • **DURGA-II** (Directionally Unrestricted Ray-Gun Array): It is a direct energy weapon developed by the Defense Research and Development Organization. • **AGNI (Ayurveda Gyan Naipunya Initiative) –** Launched by the Central Council for Research in Ayurveda Sciences (CCRAS), under the Ministry of Ayush, it serves as a platform for Ayurveda practitioners to report innovative treatments and experiences in various diseases.

Relevance: Researchers at IIT Madras have unveiled a cutting-edge tool — a detailed 3D map of five developing baby brains from the second trimester.

Subject: Science & Technology | Biotechnology

Tags: Analytical

Q77.
Answer: b
Explanation:

- **Snakebite envenoming** is a potentially life-threatening disease caused by toxins in the bite of a venomous snake. Envenoming can also be caused by having venom sprayed into the eyes by certain species of snakes that have the ability to spit venom as a defence measure.
- The World Health Organization (WHO) has classified snakebite envenoming as a **Neglected Tropical Disease (NTD)**. **So, statement 1 is correct.**
    - NTDs are communicable diseases in tropical regions, thriving in conditions of poverty and poor healthcare access. They are caused by a variety of pathogens such as viruses, bacteria, protozoa and parasitic worms. The term "neglected" reflects the lack of attention and resources despite significant impact on vulnerable communities. These diseases generally receive less funding for research and treatment than malaises like tuberculosis, HIV-AIDS and malaria.
- **In India, the 'Big Four' snakes responsible for the majority of venomous bites are:**
    - Indian cobra (*Naja naja*)
    - Common krait (*Bungarus caeruleus*)
    - Russell's viper (*Daboia russelii*)
    - Saw-scaled viper (*Echis carinatus*). **So, statement 2 is correct.**
- The Union Government has launched the **National Action Plan for Prevention and Control of Snakebite Envenoming (NAPSE)** with the aim of halving snakebite deaths by 2030. NAPSE **recommended** that snakebites should be made a notifiable disease.
- **Snakebites are not uniformly notifiable** across all Indian States. The Central Government has urged States to classify snakebites as a notifiable disease to improve data collection and response strategies. **So, statement 3 is not correct.**
    - While the **list of notifiable diseases differs from State to State** — the State governments are responsible for bringing out the notification — most of them consider infections such as tuberculosis, HIV, cholera, malaria, dengue, and hepatitis among others to be notifiable.
- A notified disease refers to a disease that is dangerous to the health of society and that requires an action by law for notifying about the disease to the Government and health authorities, who further monitor the disease and provide warning to the society.

Therefore, option (b) is the correct answer.
Relevance: The Union Health Ministry has urged states to make snakebites a notifiable disease.
Subject: Science & Technology | Health and Diseases
Tag: Factual, Conceptual

Q78.
Answer: b
Explanation:

- **The Constitution of India does not explicitly mention New Delhi** as the permanent seat of the Parliament. Instead, the Parliament usually convenes in New Delhi as per long-standing practice and convention. **So, statement 1 is not correct.**
- According to the **Article 85 of the Constitution,** the **President** has the authority to summon each House of Parliament to meet at such time and **place as deemed fit,** ensuring that **no more than six**

months elapsed between two sessions. This implies that the Parliament can be convened at any location within the territory of India, and not necessarily in New Delhi. **So, statement 2 is correct.**

- **The debate over the location** of the Parliament sessions gained attention recently when discussions arose regarding holding sessions **outside New Delhi,** prompting concerns about **logistical challenges** and constitutional validity.

Therefore, option (b) is the correct answer.
Relevance: Recently, a YSR Congress Member of Parliament (MP) suggested holding two Parliament sessions annually in South India, pointing out the logistical and climatic difficulties faced by MPs due to the harsh winters and scorching summers in Delhi.
Subject: Polity | Parliament
Tags: Conceptual

Q79.
Answer: b
Explanation:

- The President of India derives its **pardon** authority from **Article 72** of the Constitution, which allows clemency in broader cases. The President can grant pardons for:
  - Violations of Union laws.
  - Death sentences, including for state offences.
  - Matters under the Union's executive domain. **So, statement 1 is correct.**
- Additionally, **the respective Governors of the States hold similar powers under Article 161,** though they cannot pardon death sentences.
- The President of the United States can grant pardons and reprieves for federal offences but does not have the authority to pardon state offences; that power resides with state governors. **So, statement 2 is correct.**
  - Only the President of India can pardon death sentences, including those involving state laws. In the US, state governors may pardon death sentences for state crimes.
- In India, pardons are granted on the **advice of the Council of Ministers,** as required by Article 74 of the Constitution. **So, statement 3 is not correct**.
  - In the U.S., the President exercises the pardoning power independently, without requiring advice or consent from other branches of government.

Therefore, option (b) is the correct answer.
Relevance: President Joe Biden became the first US president to pardon his son, Hunter Biden, who pleaded guilty to federal tax charges.
Subject: Polity | Union Executive
Tag: Analytical

Q80.
Answer: a
Explanation:

- **Tuberculosis**: It is a **bacterial disease** caused by the bacteria *Mycobacterium tuberculosis*. As per the recent Global Tuberculosis report 2024, nearly a quarter of global case load is in India. **So, pair 1 is not correctly matched.**

- **Syphilis**: It is commonly known as the cupid's disease, because of sexual transmission being the disease's most common mode of transmission. It is caused by a **spirochaete bacterium** *Treponema pallidum*. **So, pair 2 is not correctly matched.**
- **Malaria:** As per World Malaria Report 2024, nearly 90% of all malaria deaths in the world are reported from India. ***Plasmodium vivax***, a **protist (not bacteria)** is the major malaria causing organism in India, with it contributing nearly 46% of the cases all over India. **So, pair 3 is not correctly matched.**
- **Mucormycosis**: It is a systemic **fungal infection**, which was in news because of being prevalent in patients with COVID-19. **So, pair 4 is correctly matched.**
- **Gonorrhea**: It is caused by **bacteria** *Neisseria gonorrhoeae*, which is a sexually transmitted bacteria. Gonorrhea, as per WHO, is the second most common sexually transmitted disease after Chlamydia. **So, pair 5 is correctly matched.**

So, only two of the above pairs are correctly matched.
Therefore, option (a) is the correct answer.
Relevance: Global Tuberculosis Report and World malaria report have underscored the high caseloads of these two diseases in India.
Subject: Science & Technology| Health and Diseases
Tags: Factual

Q81.
Answer: d
Explanation:

- **Article 348** of the Constitution of India states that the "**authoritative texts**, of all Bills to be introduced or amendments thereto be moved in either House of Parliament shall be in **the English language". So, statement 1 is correct.**
- **The authoritative texts include:**
  - all **Bills to be introduced or amendments** thereto to be moved in either House of Parliament or in the House or either House of the Legislature of a State.
  - all **Acts passed by Parliament or the Legislature of a State** and of all **Ordinances** promulgated by the President or the Governor of a State, and
  - all **orders, rules, regulations and bye-laws** issued under this Constitution or under any law made by Parliament or the Legislature of a State, shall be in the English language.
- **Article 348(3)** of the Constitution requires that when a Bill is introduced or passed in the Legislature of a State in a language other than English, an **English translation mus**t be provided. **So, statement 2 is correct.**
- **Article 348(1)(a)** of the Constitution specifies that all proceedings in the Supreme Court and in every High Court shall be conducted in English, unless Parliament by law otherwise provides**. So, statement 3 is correct.**
  - The **Official Languages Act, 1963,** under Section 7, permits the use of Hindi or the official language of the state in High Court proceedings, provided the Governor of the state, with the President's consent, authorizes it.

Therefore, option (d) is the correct answer.
Relevance: A new row has erupted in Parliament, with several Rajya Sabha MPs, especially from South India, taking objection to Hindi or Sanskrit titles of some recent Bills.
Subject: Polity | Languages

Tag: Conceptual

Q82.
Answer: b
Explanation:

- **Hydropower cooperation** has been an important pillar of India-Bhutan bilateral economic partnership and it has been a productive partnership for both countries.
- Indo-Bhutan hydropower cooperation began in **1961 with the signing of the Jaldhaka agreement.** The Jaldhaka project is situated on the Indian side of Indo-Bhutan border in West Bengal. The major part of power produced at Jaldhaka hydropower plant was exported to southern Bhutan.
- **Chukha Hydropower Project:** Commissioned in **1987,** the 336 MW Chukha Hydropower Project was Bhutan's **first mega power project, fully funded by the Government of India** with a **60% grant and a 40% loan**. **So, point 2 is correct.**
- **Kholongchhu Hydropower Project:** The 600 MW Kholongchhu Hydropower Project is a joint venture between **Satluj Jal Vidyut Nigam Limited of India** and Bhutan's Druk Green Power Corporation**. So, point 3 is correct.**
- **Mangdechhu Hydroelectric Project:** Commissioned in 2019, the 720 MW Mangdechhu Hydroelectric Project is among the largest joint projects between India and Bhutan with 70% loan share in funding. **So, point 5 is correct.**
- **Pancheshwar Multipurpose Project:** It is a multipurpose project developed over the Mahakali river. It is developed through a bi-partite treaty between India and Nepal. **So, point 1 is not correct.**
- **Salma Dam:** It is also known as **friendship dam,** developed by India in Afghanistan over HAri river. **So, point 4 is not correct.**

So, only three of the above given points are correct.
Therefore, option (b) is the correct answer.
Relevance: Despite the strong bilateral ties between Bhutan and India, unfavourable opinions about India's involvement in the hydropower sector are emerging.
Subject: Polity | International Relations
Tags: Factual

Q83.
Answer: a
Explanation:

- **The National Investigation Agency (NIA)** is a statutory body set up under the National Investigation Agency Act, 2008.
- **The Act extends to the whole of India and it applies also—**
  - to citizens of India outside India;
  - to persons in the service of the Government wherever they may be;
  - to persons on ships and aircrafts registered in India wherever they may be;
  - to persons who commit a Scheduled Offence beyond India against the Indian citizen or affecting the interest of India.
- **The National Investigation Agency (NIA**) is empowered to investigate and prosecute offences under various Acts, including the **Unlawful Activities (Prevention) Act, 1967**, and the Explosive Substances Act, 1908. **So, statement 1 is correct.**

- **Unlike the Central Bureau of Investigation (CBI), the NIA does not require prior consent** from state governments to take over the investigation of cases within their jurisdictions. Under Section 6(5) of the NIA Act, 2008, the Central Government can direct the NIA to investigate a scheduled offence anywhere in India without the consent of the state government concerned. **So, statement 2 is not correct.**
- The NIA is authorized to investigate **scheduled offences committed outside India**, provided such investigation is in accordance with international treaties and the domestic laws of the respective countries. **So, statement 3 is correct.**

Therefore, option (a) is the correct answer.
Relevance: The National Investigation Agency (NIA), which registered 640 cases since its inception, has recorded a conviction rate of 95.23% in 147 cases which have been decided by courts.
Subject: Polity | Constitutional and Non-constitutional bodies
Tag: Conceptual

Q84.
Answer: c
Explanation:

- A military exercise is a training event that uses military resources to simulate combat and test strategies and tactics.
- **Exercise CINBAX** is a joint **military exercise** between **India and Cambodia** which aims to war-game conduct of joint Counter Terrorism (CT) operations. **So, pair 1 is correctly matched.**
- The exercise will focus on discussions pertaining to establishment of Joint Training Task Force for Intelligence, Surveillance and Reconnaissance besides planning of operations in the CT environment.
- **Garuda Shakti** is a joint **military exercise** between **India and Indonesia** that strengthens the two countries' armed forces and military cooperation. **So, pair 2 is not correctly matched.**
- **Exercise HARIMAU SHAKTI** is an annual training event between the **Indian and Malaysian Army** which has been conducted annually since 2012. **So, pair 3 is correctly matched.**
- **SIMBEX** stands for Singapore India Maritime Bilateral Exercise, an annual **naval exercise** between the **Indian Navy and the Republic of Singapore Navy (RSN).** It was previously known as Exercise Lion King when it began in 1994. **So, pair 4 is correctly matched.**

So, only three of the above given pairs are correctly matched.
Therefore, option (c) is the correct answer.
Relevance: The latest edition of CINBAX was concluded recently in December 2024.
Subject: Polity | International Relations
Tags: Factual

Q85.
Answer: a
Explanation:

- The Amrit Gyan Kosh portal provides a shared knowledge bank with India-centric case studies. The Faculty Development portal focuses on training practitioners and faculty to improve knowledge delivery to civil servants.
- The **Amrit Gyaan Kosh Portal**, launched by the Ministry of Personnel, Public Grievances & Pensions and is the nodal agency for the portal. **So, statement 1 is not correct.**

- Developed jointly by the **Capacity Building Commission** and **Karmayogi Bharat**, the portal is hosted on the iGOT platform and serves as a comprehensive repository of best practices from across the country.
- It is developed in partnership with the **Stanford Leadership Academy for Development** and the **Asian Development Bank Institute**, the portal is aimed at fostering excellence in governance and advancing India's capacity-building ecosystem, setting the foundation for Vikasit Bharat 2047. **So, statement 2 is not correct.**
- **The initiative aims to empower public administrators** by curating resources that meet global standards while addressing India's unique administrative challenges. **So, statement 3 is correct.**

So, only one of the above given statements is correct.
Therefore, option (a) is the correct answer.
Relevance: The Amrit Gyaan Kosh Portal was recently launched in December 2024 by DoPT**.**
Subject: Polity & Governance | Miscellaneous
Tags: Factual

Q86
Answer: a
Explanation:

- ***Melye-amiley*** is a traditional fermented bamboo shoot variety from **Tripura**. It has **anti-obesity effects** and offers a solution for **weight management** and **metabolic health**.
- It reduces **lipid accumulation** and increases **fatty acid β-oxidation**. The extract boosts energy expenditure in **white adipocytes** by enhancing mitochondrial biogenesis through the **AMPK signaling pathway**.
- It upregulates **thermogenic protein expression** and increases the expression of lipolytic genes in the human body, thus aiding in fat utilization.

Therefore, option (a) is the correct answer.
Relevance: A recent study has underscored the anti-obesity properties of Melye-Amiley.
Subject: Environment | Miscellaneous(env)
Tags: Factual

Q87.
Answer: a
Explanation:

- **The Nutrient Based Subsidy (NBS)** scheme applies **only to phosphatic and potassic (P&K) fertilisers**, such as DAP (Diammonium Phosphate), MOP (Muriate of Potash), and NPK (Nitrogen, Phosphorus, and Potassium) fertilisers. However, **urea,** which is the **most commonly used fertiliser** in India, is **not covered under the NBS** scheme. Urea falls under a separate subsidy regime where its price is controlled by the government. **So, statement 1 is not correct.**
- Under the NBS scheme, the **Central Government does not fix the Maximum Retail Price (MRP)** of fertilisers. Instead, the government fixes a **per kilogram subsidy** for nutrients like nitrogen (N), phosphorus (P), potassium (K), and sulphur (S), and it allows **fertiliser companies to determine the MRP** based on market dynamics. However, for urea, the government directly controls and fixes the MRP. **So, statement 2 is not correct.**

- **The subsidy under the NBS scheme is not paid directly to farmers,** but rather to fertiliser manufacturers/importers based on the nutrient content per kilogram. This helps reduce the price for farmers, ensuring affordability. Farmers do not receive any direct subsidy amount. **So, statement 3 is not correct.**
- The **PM-PRANAM (Prime Minister Programme for Restoration, Awareness, Nourishment, and Amelioration of Mother Earth)** was introduced to incentivize states to promote the balanced use of fertilisers and r**educe chemical fertiliser consumption.** It encourages sustainable agricultural practices and organic alternatives to minimise environmental degradation. **So, statement 4 is correct.**

Therefore, option (a) is the correct answer.

Relevance: Cabinet approves extension of One-time Special Package on Di-Ammonium Phosphate (DAP) beyond the NBS subsidy w.e.f 01.01.2025 to ensure sustainable availability of DAP at affordable prices to the farmers.

Subject: Current Affairs | Economics | Agriculture

Tags: Factual

Q88.

Answer: c

Explanation:

- The Union Cabinet recently approved the Indian government's 'One Nation, One Subscription' (ONOS) scheme.
- **The Information and Library Network (INFLIBNET)** will oversee the implementation of the ONOS scheme. **So, statement 1 is correct.**
- **ONOS objective is to provide equitable access to research articles,** regardless of an institution's prestige or financial capacity. **So, statement 2 is correct.**
- The ONOS entail, the first mooted around 2018-2019, the scheme's ambitious rollout comes with a substantial financial outlay of ₹6,000 crore over three years (2025-2027), to be paid to 30 major international journal publishers.
- The Indian public and academic institutions currently spend approximately ₹,500 crore annually on journal subscriptions.
- This estimate may include database subscription costs; the total public expense for journal access is likely under ₹7,000 crore per year.
- ONOS aims to democratize knowledge but involves complexities that require deeper analysis.

Therefore, option (c) is the correct answer.

Relevance: The Union Cabinet recently approved the Indian government's 'One Nation, One Subscription' (ONOS) scheme.

Subject: Polity & Governance | Miscellaneous

Tags: Factual

Q89.

Answer: c

Explanation:

- The **Captive Elephant (Transfer or Transport) Rules, 2024**, were introduced to regulate the movement of captive elephants within and between states. **So, statement 1 is correct.**
- **The Ministry of Environment, Forest and Climate Change (MoEF&CC) developed the Gajah Suchana mobile app** in collaboration with the Wildlife Institute of India for DNA profiling of

captive elephants, creating a genetic database to validate their identity and combat illegal transfers. **So, statement 2 is correct.**

- Elephants are listed in **Schedule I** of the Wildlife (Protection) Act, 1972, receiving the **highest legal protection**.
- Despite this, the elephant is the **only animal in Schedule I** that can still be legally owned through **inheritance or as a gift**. **So, statement 3 is correct.**
- Section 3 of the WPA (2003) prohibited trade in captive wildlife and restricted non-commercial transfers across state boundaries without permission from the **Chief Wildlife Warden**.

So, all of the statements given above are correct.
Therefore, option (c) is the correct answer.
Relevance: Kerala Government recently issued permits for transporting elephants for religious festivals.
Subject: Environment | Biodiversity and Conservation
Tags: Conceptual

Q90.
Answer: d
Explanation:

- Governments around the world traditionally impose a "sin tax" on products that negatively impact public health, such as alcohol and tobacco. These goods are taxed at higher rates to achieve two primary goals: discouraging their consumption and using the revenue generated to fund welfare programs.
- The Goods and services tax regime recognizes sin goods and has provisions for higher taxationsThese goods are placed under the highest **GST tax slab of 28%**, with an **additional compensation cess** that varies between **11% and 290%,** depending on the product category. The cess is imposed to **discourage the consumption** of such goods while generating **additional revenue**.
- As of now, the following categories of products are taken to as sin goods:
    - **Aerated Beverages. So, point 2 is correct.**
    - **Cigarettes** and other tobacco products. **So, point 1 is correct.**
    - **Pan Masala. So, point 4 is correct.**
    - Caffeinated drinks
    - Sugar
    - **Gambling** and Lottery. **So, point 3 is correct.**
    - **High end fossil fuel based automobiles. So, point 5 is correct.**

So, all of the above are considered as sin goods.
Therefore, option (d) is the correct answer.
Relevance: The Group of Ministers (GoM) on GST rate rationalization has proposed an increase in the tax rate on sin goods, including aerated beverages, cigarettes, tobacco, and related products, from 28% to 35%.
Subject: Economics | Taxation
Tags: Factual

Q91.
Answer: b
Explanation:

- Manual scavenging is **banned under the Prohibition of Employment as Manual Scavengers and their Rehabilitation Act, 2013 (PEMSR)**.

- The Act prohibits employing individuals to manually clean, carry, dispose of, or handle human excreta until its disposal. **So, statement 2 is correct.**
- The Act provides for **an explicit definition for the practice of manual scavenging** which also includes people employed to clean septic tanks, ditches, or railway tracks. **So, statement 1 is not correct.**
- The Act recognizes manual scavenging as a "dehumanizing practice" and highlights the need to "correct the historical injustice and indignity suffered by the manual scavengers."
- The **National Action for Mechanised Sanitation Ecosystem (NAMASTE) scheme** is a program that aims to improve the safety and dignity of sanitation workers in India. **So, statement 3 is correct.**

So, only two of the above given statements are correct.
Therefore, option (b) is the correct answer.
Relevance: A recent study has underscored the prevalence of caste dominance amongst manual scavengers.
Subject: Polity | Governance
Tags: Factual

Q92.
Answer: a
Explanation:

- **Hemophilia,** which means love (*philia*) of blood (*hemo*), is the most common severe hereditary hemorrhagic disorder.
- The patients suffering from this disorder present with **prolonged clotting time**, that is they **bleed more profusely than average people.** This is because the clotting of blood is dependent on a set of certain molecules named as clotting factors. Persons suffering from this disorder are deficient in one or the other clotting factors. **So, statement 1 is not correct.**
- It is an **autosomal genetic disease**, that is it transmits from one generation to others via the sex chromosomes. **So, statement 2 is correct.**
- It is **transmitted via the X chromosomes**; human females have a pair of X chromosomes, while males have just one X chromosome. Even if one of the X chromosomes is defected, women will not suffer from Hemophilia because of the presence of a healthy X chromosome. But in case of males if the one X chromosome is affected, it results in the disease. That is why males **are more affected than females**. **So, statement 3 is correct.**

So, only one of the above given statements is not correct.
Therefore, option (a) is the answer.
Relevance: Indian scientists have developed a novel gene therapy for Hemophilia
Subject: Science & Technology| Health and Diseases
Tags: Factual

Q93.
Answer: d
Explanation:

- **Land degradation** is a process where **land becomes less healthy and productive** due to a combination of human activities or natural conditions. It reduces agricultural yields by **diminishing soil fertility,** leading to a l**oss of essential nutrients** like nitrogen, phosphorus, and potassium, which are crucial for plant growth, ultimately resulting in **smaller crop harvests** and **lower productivity. So, point 1 is correct.**

- Land degradation can lead to **changes in rainfall patterns** primarily by reducing the amount of vegetation cover, which in turn decreases the amount of **water vapor transpired back into the atmosphere**, leading to less cloud formation and ultimately, reduced rainfall. **So, point 2 is correct.**
- Land degradation disrupts the natural soil condition contributing to **nitrous oxide emissions,** leading to increased availability of nitrogen in forms that can be readily converted into nitrous oxide by **soil microbes,** particularly through processes like **denitrification**, which occurs when soil becomes poorly aerated due to factors like **compaction, erosion, and reduced organic matter** content caused by degradation; this results in higher nitrous oxide release into the atmosphere. **So, point 3 is correct.**
- Land degradation leads to **changes in surface albedo** by **replacing dark, vegetation-covered surfaces** with lighter, bare soil or sparsely vegetated areas, which **reflect more sunlight back into space,** thereby increasing the overall albedo of the affected region. **So, point 4 is correct.**
- Land degradation, particularly through **soil erosion,** leads to **reduced soil fertility** by physically **removing the topsoil layer** which contains the **majority of essential nutrients** needed for plant growth, resulting in a decline in crop yields and a less productive land area. **So, point 5 is correct.**

Therefore, option (d) is the correct answer.
Relevance: Recently, the 16th Conference of the Parties (COP16) to the United Nations Convention to Combat Desertification (UNCCD) concluded in Riyadh, Saudi Arabia.
Subject: Environment | Climate Change
Tags: Factual

Q94.
Answer: a
Explanation:

- Different types of clouds have different effects on the Earth's climate. While some types of clouds help to warm the Earth, others help to cool it.
- **Cirrus clouds are cold clouds that form high in the troposphere.** Cirrus clouds trap heat from getting out of the Earth system. Therefore, they have a net warming effect**. A warming Earth will likely produce more cirrus clouds**, potentially resulting in a self-reinforcing loop.
- As the lower atmosphere warms up and the upper atmosphere cools down, it will allow buoyant clouds to rise higher in the sky. These clouds would trap heat and this would result in considerable warming of the atmosphere. **So, statement 1 is correct.**
- An increase in these clouds acts as a blanket that traps heat, exacerbating global warming. Conversely, a decrease in cirrus clouds allows heat to escape from the atmosphere.
- Cirrus clouds warm the earth, and thus thinning or removing these clouds would reduce their heat trapping capacity, resulting in a cooling effect on Earth's climate. These clouds reflect little sunlight, and they are so cold that they absorb most of the thermal radiation that comes their way. **Hence they warm more than they cool. So, statement 2 is not correct.**
- According to recent studies, **cirrus clouds formation will likely amplify climate change in the future. This is called positive feedback.** For example, if a warming climate causes more clouds that trap heat, this can cause even more warming. Positive, in this case, does not mean "good," it means that the problem of climate warming is compounded. **So, statement 3 is not correct.**
- Fewer bright, low clouds mean the planet "has darkened," allowing it to absorb more sunlight. Low clouds also feed into this effect as they reflect away sunlight.

- Low-level clouds tend to thrive in a cool and moist lower atmosphere. As the planet's surface heats up, this can cause them to thin or dissipate entirely, setting up a complicated feedback loop where low clouds are disappearing because of global warming, and their disappearance then drives further warming.

Therefore, option (a) is the correct answer.
Subject: Geography | Climate Weather and Atmosphere
Relevance: Increasing global temperatures are contributing to increased formation of cirrus clouds, reinforcing global warming
Tags: Conceptual

Q95.
Answer: d
Explanation:

- AgriStack is a Digital Public Infrastructure (DPI) designed to modernize agriculture in India. Announced in the Union Budget, it aims to bring various stakeholders—farmers, government agencies, agritech firms, and financial institutions—onto a single digital platform to enhance productivity and decision-making.
- AgriStack creates a comprehensive repository of farmer-related information, including land records, crop data, insurance coverage, income history, and more. A unique **digital ID linked to Aadhaar** will be assigned to farmers to streamline access to schemes, formalize land records, and ensure targeted delivery of benefits. **So, statement 1 is not correct.**
- Another component of the Digital Public Infrastructure (DPI), the Krishi Decision Support System, is a geo-spatial platform for assessing the cropping patterns in the region.
- AgriStack will be implemented by the **Ministry of Agriculture and Farmers Welfare** in collaboration with state revenue and agriculture departments. While **ISRO** contributes satellite data and technological expertise, the primary responsibility lies with the Agriculture Ministry, not MeitY. **So, statement 2 is not correct.**
- AgriStack's formalization of land records and creation of digital farmer identities will benefit **over six crore farmers**. The initiative aims to improve service delivery, access to credit, and the effectiveness of government schemes, thus uplifting the livelihoods of Indian farmers.

Therefore, option (d) is the correct answer.
Subject: Economics | Economic Schemes and Policies
Tags: Factual

Q96.
Answer: a
Explanation:

- High interest rates increase the cost of borrowing, which can reduce consumer spending and business investments. A **tight monetary policy**, characterized by higher interest rates, aims to reduce inflationary pressures but can also slow down economic growth by discouraging borrowing. **So, statement 1 is correct.**
- The **credit-deposit ratio** indicates how much of the bank's deposits are being lent out as loans. A higher ratio suggests that banks are actively lending, which can stimulate economic activity by providing more credit to businesses and consumers. It reflects the efficiency of the banking sector in channeling savings into productive investment. **So, statement 2 is correct.**
- Monetary policy mainly influences **aggregate demand** and overall price levels through interest rates, not directly agricultural production. While monetary policy can influence inflation, it doesn't directly

address **food inflation**, which is often driven by factors like supply-side shocks (e.g., poor harvests, logistical issues) and government policies on food procurement. **So, statement 3 is not correct.**

Therefore, option (a) is the correct answer.
Subject: Economics | Basic Economic Concepts
Tags: Conceptual

Q97.
Answer: a
Explanation:

- A dam is a structure built across a river or stream to hold back water. These are constructed for the purpose of storing water, generating hydroelectricity, flood control etc.
- In recent times studies have underscored some ecological impacts of these dams, foremost the construction of dams **increases the seismicity of the region**, as strong currents from the stream push against the dam wall leading to creation of tectonic shockwaves in the region. Further, the dam creation requires deep drilling of soil, predisposing the region for leaching of water deeper to tectonic plates leading to generation of **earthquake swarms,** via a phenomenon known as **hydroseismicity**. **So, statement 1 is correct.**
- The recent event of landslide in Wayanad is increasingly linked to the century old Mullaperiyar dam, as construction of dam disrupts the natural hydrological flow in the region, also because of increased water storage **solifluction can take place**, thus triggering landslides. **So, statement 2 is correct.**

Therefore, option (a) is the correct answer.
Relevance: Studies have underscored the linkage between Mullaperiyar dam and Wayanad landslides.
Subject: Geography | Miscellaneous(geo)
Tags: Conceptual

Q98.
Answer: c
Explanation:

- In 1944, the **All India Hindu Mahasabha** adopted the **Constitution of the Hindustan Free State (CHFS)** as a possible draft constitution. The Mahasabha was at that time led by **Vinayak Damodar Savarkar. So, statement 1 is correct.**
- It was written by DV Gokhale, Laxman Bhopatkar, KV Kelkar and MR Dharmadhere, with a foreword by N.C. Kelkar.
- The CHFS provided that independent India would be a **democratic state with no state religion. So, statement 2 is correct.**
- The framers of the CHFS believed that democracy in India had evolved to such a stage that people were entitled to "**a direct voice in respect of their social and economic well-being and political destiny.**" It even made provisions for instruments of **direct democracy – Initiative, Recall** etc. **So, statement 3 is correct.**
- The framers of the document chose the name Free state of Hindoostan for the nation against the popular names India or Bharat.

So, all of the above statements are correct.
Therefore, option (c) is the correct answer.
Relevance: Year 2024 marked the 75th year of the Indian constitution.
Subject: History | Partition and Independence

Tags: Factual

Q99.
Answer: b
Explanation:

- **Regulation and development** of **oilfields and mineral oil resources; petroleum and petroleum products**; other liquids and substances declared by Parliament by law to be dangerously inflammable fall under **Entry 53 of the Union List (Seventh Schedule of the Constitution).** This gives Parliament exclusive power to legislate on these matters. **So, statement 1 is not correct.**
- In India, **mineral oils** refer to naturally occurring hydrocarbons such as **petroleum and natural gas,** which are crucial for energy production. **The Oilfields (Regulation and Development) Amendment Bill, 2024**, expands this definition to include **coal bed methane** and **shale gas/oil,** while explicitly excluding coal, lignite, and helium. **So, statement 2 is correct.**

Therefore, option (b) is the correct answer.
Relevance: The Rajya Sabha recently passed the Oilfields (Regulation and Development) Amendment Bill, 2024, which amends the Oilfields (Regulation and Development) Act, 1948.
Subject: Economics | Infrastructure
Tags: Factual

Q100.
Answer: c
Explanation:

- **The seating arrangement in the Lok Sabha** is organized in a way that reflects the **party's strength in the house.** The ruling party and the opposition are seated on opposite sides of the chamber, with the **party with the largest number of seats** occupying the **treasury benches.** The number of **seats allocated to a party** is proportionate to its representation in the house. This practice is followed to maintain **order and decorum** in the house. **So, statement 1 is correct.**
- The specific **allocation of seats to individual members** is generally done by the **party leadership** in consultation with the Lok Sabha Secretariat. **Party whips or floor leaders** play a crucial role in seat assignments based on factors such as **seniority, portfolio, and prominence** within the party. This ensures **efficient functioning** of the house and coordinated decision-making during debates and voting. **So, statement 2 is correct.**
- The first seat in the front row on the right, irrespective of Lok Sabha membership, goes to the Prime Minister. Ministers are seated in the front rows on the right, in order of seniority. Members must occupy their allotted seats and address the House from them unless directed otherwise by the Chair. **So, statement 3 is correct.**

So, all three of the given statements are correct.
Therefore, option (c) is the correct answer.
Relevance: Members of Parliament from both NDA and Opposition parties have expressed dissatisfaction with the Lok Sabha seating arrangement, feeling that it does not reflect their seniority or political stature
Subject: Polity | Parliament
Tags: Factual

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